REDSTONE CSP PROJECT

Project title	Redstone CSP Project
Project ID	5146
Crediting period	25-October-2024 to 24-October-2031
Original date of issue	12-August-2024
Most recent date of issue	18-August-2024
Version	1.5
VCS Standard Version	4.7
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1 PROJECT DETAILS

1.1 Summary Description of the Project

Redstone concentrated solar power (CSP) project is a nominal 100MW integrated concentrated solar thermal energy plant constructed on the farm 469 the Hay RD, between the towns of Daniëlskuil and Postmasburg, in the Northern Cape Province of South Africa. It forms part of the South African Renewable Energy Independent Power Producer Procurement Programme (REIPPP). As such, it is currently one of the largest investments in South Africa under the REIPPP. The project utilizes Therma Vault technology which combines solar thermal and molten salt energy storage technologies. This enables the plant to store thermal energy for a period of up to 12 hours, and hence it can provide electricity to the equivalent of more than 200,000 South African homes during peak demand periods via the national grid, including after the sun has set.

Solar energy is captured via a circular heliostat field made up of 41,260 heliostats (each with 25 m^2 mirror reflective surface area). The heliostats track the sun and reflect the sunlight onto a 250 m tall central receiver tower, which uses molten salt as heat transfer and storage medium. Thermal energy is converted to electricity through 115MW high temperature subcritical steam turbine generating units.

The project has a 20-year power purchase agreement with the South African national grid operator (Eskom) to supply approximately 500 000 MWh of renewable energy per year starting in 2024. This CSP plant will reduce greenhouse gas (GHG) emissions by replacing carbon intensive energy with clean renewable electricity.

Prior to implementing this project electricity was supplied by the current fleet of power plants owned and operated by Eskom, the majority of which are coal-based. The Renewable Energy Independent Power Producer Procurement Programme (REIPPPP) is an initiative by the South African government aimed at increasing electricity capacity through private sector investment in solar photovoltaic and concentrated solar, onshore wind power, small hydro, landfill gas, biomass, and biogas [1]. Since coal is an emissions-intensive energy carrier, South Africa's economy is very emissions-intensive, especially when compared to other countries at a similar stage of development as measured by the Human Development Index [2]. South Africa's energy sector is by far the largest contributor to the country's high emissions, contributing 94.7% to the county's predominant GHG, which is CO_2 [3].

The project is expected to result in an average annual reduction of approximately 500 000 tCO_2e in GHG emissions and a total of 3 500 000 tCO_2e emission reductions over the first crediting period of seven years.

1.2 Audit History

Audit type	Period	Program	Validation/verification body name	Number of years
Validation	Initial Validation October 2024	VCS	Carbon Check (India) Private Ltd.	To be updated later

1.3 Sectoral Scope and Project Type

Sectoral scope[1]	Sectoral scope 01: Energy (renewable/non-renewable)
Project activity type	Project

1.4 Project Eligibility

1.4.1 General eligibility

As per paragraph 2.1.1 of the VCS Standard, version 4.7, the project is eligible under the scope of the VCS Program because this project activity meets three of the listed points, as described in *Table 1* below. The project is also eligible under VCS v4.7, table 2.1, as shown in *Table 2* below:

Table 1: Scope of the VCS program

Elements	Justification of project applicability
1) The seven Kyoto Protocol greenhouse gases.	The project aims at reducing greenhouse gas (GHG) emissions by replacing coal-based energy supplied by the South African national grid, with clean renewable electricity.
2) Ozone-depleting substances (ODS).	Not applicable
3) Project activities supported by a methodology approved under the VCS Program through the methodology development and review process.	Project activity applies the CDM ACM0002: Grid-connected electricity generation from renewable

[1] *Projects, activities, or methodologies may be developed under any of the 16 VCS sectoral scopes: https://verra.org/programs/verified-carbon-standard/vcs-program-details/#sectoral-scopes*

	sources (Version 22.0)[2] methodology, which has been approved for use under the VCS Program.
4) Project activities supported by a methodology approved under an approved GHG program, unless explicitly excluded (see the Verra website for exclusions).	Project activity applies VCS Program approved CDM methodology ACM0002: Grid-connected electricity generation from renewable sources (Version 22.0)[3].
5) Jurisdictional REDD+ programs and nested REDD+ projects as set out in the Jurisdictional and Nested REDD+ (JNR) Requirements.	N/A

Table 2: Excluded Project Activities under the VCS v 4.7 table 1

Exclusions from VCS Program Scope	Applicability
Grid-connected electricity generation activities using hydroelectric power plants. The exclusion does not apply to ocean energy (e.g., wave, tidal, salinity gradient, and ocean thermal energy conversion). For hydro projects, large scale means a maximum capacity of greater than 15MW, where maximum capacity is the installed/rated capacity or authorised capacity (as determined in the activity approval from the project regulator, government, or similar entity), whichever is lower. Grid-connected means >50% of total generation is exported to a national or regional grid.	Not applicable, this project is not a hydroelectric power plant.
Grid-connected electricity generation activities using wind, geothermal, or solar photovoltaic (PV) power plants. The exclusion does not apply to concentrated solar thermal-to-electricity, floating solar PV, or energy storage systems (e.g., batteries). Grid-connected means >50% of total generation is exported to a national or regional grid. See the VCS Program Definitions for the full definition.	The project is a concentrated solar thermal-to-electricity project, which is not affected by this exclusion, therefore this project is eligible.
Activities recovering waste heat for combined cycle electricity generation, or to heat/cool via cogeneration or trigeneration.	Not applicable, this project is not a waste heat recovery project.

[2] *https://cdm.unfccc.int/methodologies/DB/XB1TX7TAZ6SLWM9B7BC67THHVD16JV*

[3] *https://cdm.unfccc.int/methodologies/DB/XB1TX7TAZ6SLWM9B7BC67THHVD16JV*

Exclusions from VCS Program Scope	Applicability
The exclusion does not apply to waste gas recovery or electricity generation using waste heat recovery outside of combined cycle applications (e.g., organic Rankine cycles).	
Activities generating electricity and/or thermal energy for industrial use from the combustion of non-renewable biomass, agro-residue biomass, or forest residue biomass. The exclusion does not apply to gasification, pyrolysis, combusting biofuels, biogas, fractions of renewable biomass in refuse-derived fuels, agro/forest biomass residues in waste streams that are sent to landfills, CO_2 capture and storage from renewable biomass combustion, or thermal efficiency improvements (e.g., cook stoves).	Not applicable, this project does not combust biomass.
Activities generating electricity and/or thermal energy using fossil fuels, and activities that involve switching from a higher to a lower carbon content fossil fuel. The exclusion does not apply to the use of captured flare and/or vent gas or waste containing previously used petroleum products (e.g., used plastics, oils, lubricants).	Not applicable, this project does not combust fossil fuel.
Activities replacing electric lighting with more energy-efficient electric lighting, such as the replacement of incandescent electrical bulbs with compact fluorescent lights (CFLs) or light emitting diodes (LEDs). Large-scale means energy-efficient improvements with a maximum savings greater than 60 GWh/year or emission reduction greater than 60 kt CO_2e per year.	Not applicable, this project does not replace electric lighting.
Activities installing and/or replacing electricity transmission lines and/or energy-efficient transformers. Large-scale means energy-efficient improvements with a maximum savings greater than 60 GWh/year or emission reduction greater than 60 kt CO_2e per year.	Not applicable, this project does not install or replace electricity transmission lines or transformers
Activities that reduce hydrofluorocarbon-23 (HFC-23) emissions	Not applicable, this project does not reduce HFC-23 emissions.

Pipeline listing and validation deadlines - The project activity was listed before validation process began. The opening meeting with the VVB will only take place once the project activity is shown as status 'Under Validation' on the VCS Program project pipeline'. Further, validation shall not begin until the 30-day public comment period is underway, and the VVB shall not complete validation until after the 30-day public comment period has ended, and any comments received are incorporated into the VCS Project Description. As per paragraph 3.8.1 of VCS standard V4.7, non-AFOLU projects shall complete validation within two years of the project start date and this condition will be fulfilled as the start date of the project activity is anticipated to be 25 October 2024, and validation will commence in 2024.

Project activity applies CDM methodology ACM0002: Grid-connected electricity generation from renewable sources (Version 22.0), which has been approved under the VCS Programme, by Verra for project activities seeking registration with them. ACM0002 is a large-scale methodology that has no capacity limit, and the Redstone CSP plant is a stand-alone facility, so is not a fragmented part of a larger project or activity.

1.4.2 AFOLU project eligibility

This section is not applicable, seeing that this project is not an AFOLU project.

1.4.3 Transfer project eligibility

This section is not applicable, seeing that this project is not a transfer project or CPA seeking registration.

1.5 Project Design

The project has been designed as:

☒ Single location or installation

☐ Multiple locations or project activity instances (but not a grouped project)

☐ Grouped project

1.5.1 Grouped project design

This section is not applicable, seeing that this is a single location project.

1.6 Project Proponent

Organization name	ACWA Power Solar Reserve Redstone Solar Thermal Power Plant (RF) (Pty) Ltd ('Redstone')
Contact person	Nandkumar Bhula
Title	Country Manager: ACWA Power
Address	Office XX07001, 90 Grayston Drive Sandton Johannesburg 2196, South Africa
Telephone	+27 11 722 4101
Email	nbhula@acwapower.com

Organization name	Anthesis B.V. ('Anthesis')
Contact person	Grant Little
Title	Principal Carbon Developer
Address	Workshop 17, 32 Kloof Street, Cape Town, South Africa
Telephone	+27 21 300 6770
Email	grant.little@anthesisgroup.com

1.7 Other Entities Involved in the Project

There are no other entities involved in the development of this project.

1.8 Ownership

ACWA Power Solar Reserve Redstone Solar Thermal Power Plant (RF) (Pty) Ltd has the legal right to own and operate the Redstone CSP facility, which can be verified through the following documents provided by the project proponent:

- IPP license
- Lease direct agreement dated 06 December 2019
- Formalities certificate – Landowner

Anthesis and ACWA Power Solar Reserve Redstone Solar Thermal Power Plant (RF) (Pty) Ltd have a legal agreement in place for the joint ownership of a registered carbon project.

1.9 Project Start Date

Project start date	25-October-2024
Justification	According to Section 3.8 Project Start Date in VCS Standard Version 4.7, the project start date of a non-AFOLU project is the date on which the project began generating GHG emission reductions or carbon dioxide removals. In this case the start date of the Redstone CSP project activity is the Commercial Operation Date (COD), on which the plant first starts to feed electricity into the national grid. As such, that is the day on which it started generating emission reduction credits.

1.10 Project Crediting Period

Crediting period	☒ *Seven years, twice renewable* ☐ *Ten years, fixed* ☐ Other (state the selected crediting period and justify how it conforms with the VCS Program requirements)
Start and end date of first or fixed crediting period	25-October-2024 to 24-October-2031

1.11 Project Scale and Estimated GHG Emission Reductions or Removals

☐ < 300,000 tCO2e/year (project)

☒ ≥ 300,000 tCO2e/year (large project)

Calendar year of crediting period	Estimated GHG emission reductions or removals (tCO_2e)
25-October-2024 to 31-December-2024	125 000
01-January-2025 to 31-December-2025	500 000
01-January-2026 to 31-December-2026	500 000
01-January-2027 to 31-December-2027	500 000
01-January-2028 to 31-December-2028	500 000
01-January-2029 to 31-December-2029	500 000
01-January-2030 to 31-December-2030	500 000
01-January-2031 to 24-October-2031	375 000
Total estimated ERRs during the first or fixed crediting period	3 500 000
Total number of years	7
Average annual ERRs	500 000

1.12 Description of the Project Activity

Project Design

The Redstone CSP Plant project entails the construction and operation of a concentrating solar thermal power plant with associated infrastructure and services for the generation of renewable electricity, to supply to the national power grid. The project is designed to produce approximately 480 000 – 500 000 gigawatt-hours (GWh) net of renewable electricity annually, with a nominal net generating capacity of approximately 100 megawatts (MW). It is envisaged that the CSP plant will be operated as a mid-merit or base load plant and the project has entered into a 20-year power purchase agreement with the South African National grid operator (Eskom). Total construction and development costs of the plant are estimated at R6.5 billion.

The plant's auxiliary electricity consumption is 25 000 MWhe per year and it will consume approximately 200 000 m^3 of raw water per year. The project utilises ThermaVault technology, which combines solar thermal and molten salt energy storage technologies. This enables the plant to store thermal energy for up to 12 hours. The plant's heliostats track the sun and reflect the sunlight onto a 250 m tall central receiver tower, which uses molten salt as a heat transfer and storage medium. Key plant parameters are outlined below:

- Plant commissioning: 2024
- Plant lifetime: 25 years
- Planned plant availability: 98%. Unplanned availability of 94% - 96% is expected.
- Cleanliness factor: 98% (percentage of reflective area not affected by dust fouling)
- Load factor: The Redstone CSP Plant is expected to operate at full load or partial loads depending on the solar resource throughout its operating life. The unit shall start-up and shut down daily.
- Expected net generation: 496,596 GWh per annum

The following auxiliary infrastructure forms part of the project:

- **Solar collector field:** Consisting of approximately 41 260 dual-axis (azimuth and elevation), tracking heliostats, each with a surface area of 25m^2. The solar resource is estimated at 2 709 kWh/m^2, and the heliostats operate at a reflectivity of 93%.
- **Receiver:** A 250-meter-tall concrete tower and thermal receiver rated at 600 MW thermal (MWth). The radiation from the solar field is reflected onto the receiver and converted into thermal energy, heating the molten salts from 300ºC to 565ºC. The molten salts are a mixture of nitrate salts (60% $NaNO_3$ and KNO_3) which remain in liquid state in the working temperature range of the Receiver. The receiver will operate at an absorptivity of 96,9%, and an availability of 98,56% is planned.
- **Cold and hot storage tanks***:* The *cold salt storage tank* holds the cold molten salts (300ºC), from where it is pumped to the receiver where the salt is heated to 565 ºC. From the receiver, the hot salt descends down the tower to the *hot salt storage tank.* The molten salt storage system enables the plant to maximize power generation during peak demand periods and to continue generating electricity after sunset. The storage capacity is 2 950 MWth, which provides 12 hours of storage. Efficient thermal insulation on the tanks and secondary heating systems mitigates the potential for solidification of the molten salt medium.
- **Steam generation system (SGS):** From the hot salt tank, the molten salt is pumped with a circulation pump to the steam generator, which produces high pressure superheated steam in the steam generation train (preheater- evaporator-superheater). The steam is then sent to the high-pressure body of the steam turbine, where its first expansion is produced.
- **Power block:** A thermal to electric power block, with a 115 MWe gross power and 100Mwe net power at design conditions, transforms the thermal energy to kinetic energy

or mechanical energy of rotation and then into electrical energy. The steam generator system consists of a high-pressure turbine (HPT), an intermediate reheater, and a low-pressure turbine (LPT) with a condensing exhaust. The mechanical energy is transferred from the steam to the rotation of the turbine and then to the alternator, it is transferred through the generator to the main transformer, which will raise the power output. The turbine efficiency at 29.5˚C is 46,83 %, and availability of 96,86 % is expected.

- **Cooling tower**: An air-cooled condenser and/or a cooling tower for the steam cycle is in place to minimise the consumption of water.
- **Water reticulation and purification works:** Water is obtained from the Sedibeng Bulk Water Supply Pipeline for industrial water use and is demineralised water for the boiler system. The water treatment plant includes the following systems:
    - Ultra filtration
    - Reverse osmosis
    - Electro deionization (EDI) System
    - Chemical dosing
- **Sewer reticulation and treatment works** for treating wastewater generated on-site.
- **Effluent treatment system:** This treats effluent from the power plant and comprises the following systems:
    - Effluent collection and equalization subsystem
    - Effluent control, conditioning and discharge subsystem
    - Blowdown equalization and conditioning subsystem
    - An evaporation pond consisting of three compartments with a combined area of approximately 8.0 ha.
- **Associated Infrastructure:**
    - Roads and storm water infrastructure
    - Two Emergency diesel generators
    - Two liquid gas or diesel auxiliary burners for start-up
    - Substation and switchyard of approximately 100 m x 100 m containing step-up transformers and associated structure. The connection shall be at a voltage level of 132 kV injected into the Eskom distribution system.
    - Approximately 8km overhead power lines connecting to the Eskom grid

- Dry cooling system for the power generation cycle
- Administrative and office buildings
- Visitors' centre
- Equipment and materials lay down area
- Assembly Plant (during construction phases)
- Concrete batching plant (during construction phases)
- Vehicle workshops and wash bays
- Fuel storage area
- Temporary general waste storage facility
- Hazardous material storage facility.

A diagram of the process is shown in *Figure 1*, and *Figure 2* to *Figure 5* show the Redstone project during the construction phase.

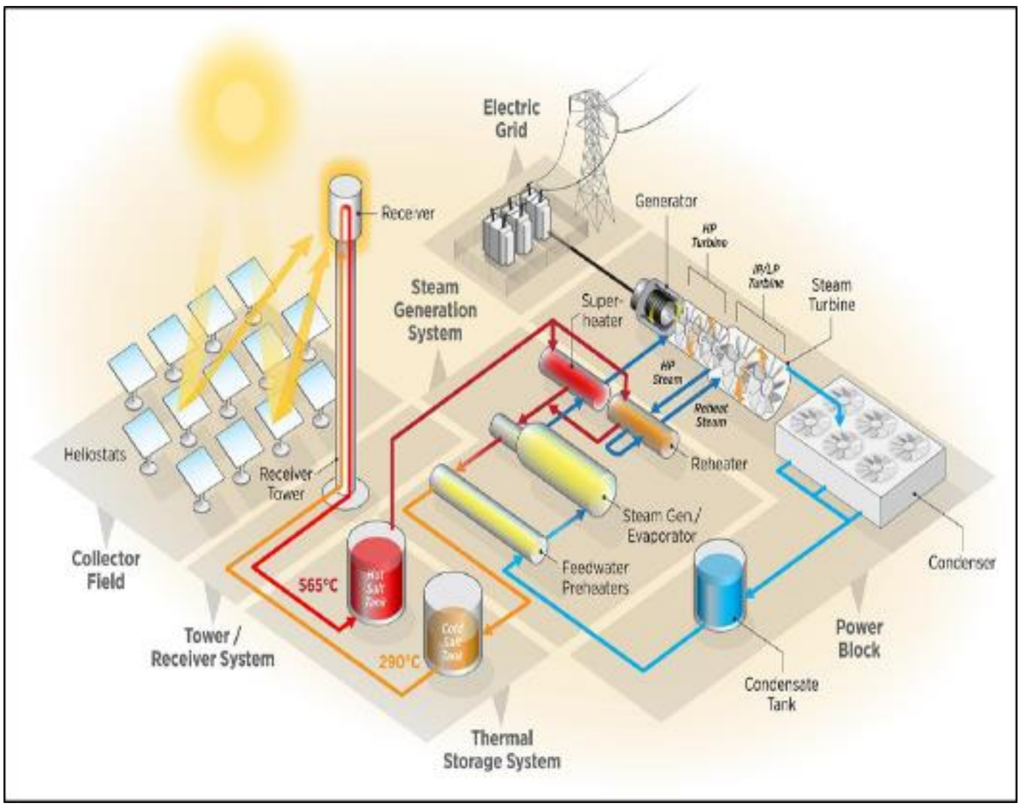

Figure 1: Diagram of a Concentrated Solar Plant

Figure 2: Redstone Solar Receiver Tower (August 2024)

Figure 3: Heliostats being installed (August 2024)

Figure 4 Redstone receiver tower under construction (2022)

Figure 5: Aerial view of Redstone CSP Plant (foreground), with Jasper and Lesedi Solar PV plants (in the background).

<u>GHG Emission reductions</u>

Fossil-fuel based power generation, including electricity, is one of the largest sources of CO_2 emissions globally *[4]*. In South Africa, the dominant gas contributing most to the country's GHG emissions is CO_2 (83.6% in 2020). The energy sector is by far the largest contributor to CO_2 emissions in South Africa, contributing 81% of total GHG emissions in 2020 *[3]*. A significant contributing factor is South Africa's reliance on fossil fuel-based electricity production (produced from coal, diesel and fuel oil) *[5]*. This project will reduce GHG emissions by replacing fossil fuel-intensive electricity fed into South Africa's national grid with clean renewable electricity.

<u>Implementation Schedule</u>

The project implementation schedule is shown in Table 3 below.

Table 3: Project Implementation Schedule

NO.	Activity	Optimized Start Date	Optimized Finish Date
1	Feed Water and Recirculation System	May/24	Jul/24
2	TURBINE SYSTEM	Jul/24	Jul/24
3	PLANT START UP	Aug/24	Sep/24
4	SOLAR FIELD SYSTEM	Jun/24	Feb/25
5	COD Between Redstone and Eskom	Aug/24	Oct/24

1.13 Project Location

The project site is located on the remainder of the Farm 469, the Hay District (Administration District), approximately 5 km southeast of the Groenwater community and 30 km east of Postmasburg and falls within the jurisdiction of the Tsantsabane Local Municipality of the ZF Mcgawu District in the Northern Cape Province, South Africa. The Project Site is accessed via the existing regional road, R385 (also called TR70/1), located at the northern boundary of the Plant.

The Redstone CSP plant is located at the following GPS coordinates:

Latitude: 28°17'31.95"S / Longitude: 23°22'21.17"E

The locality map is shown in Figure 6 below.

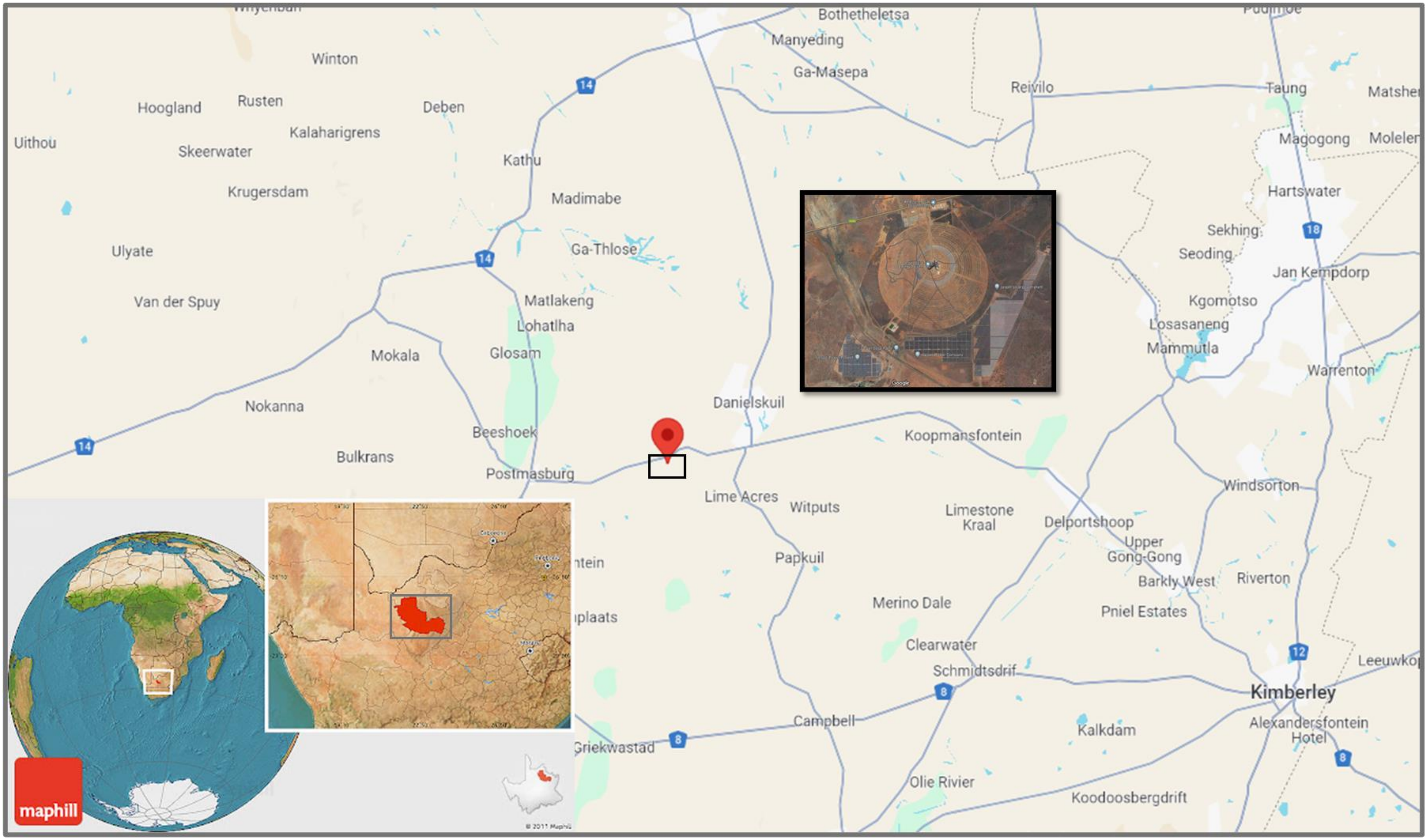

Figure 6: Locality Map of Redstone CSP plant

1.14 Conditions Prior to Project Initiation

The electricity generation conditions in South Africa, prior to including this CSP plant into the national grid, are described under Section 3.4 (Baseline Scenario).

Although this is not an AFOLU project, the environmental conditions are discussed as this is a Greenfields project and is established on an area previously zoned as agricultural land. The primary purpose of the project is to supply renewable and reliable electricity to South African consumers, and as such, it is not implemented with the aim of generating GHG emissions for the purpose of their subsequent reduction, removal, or destruction. The conditions existing prior to initiation of the project are described below.

Ecosystem type:

The CSP facility is situated within the Savanna Biome, the largest Biome in southern Africa. The Kalahari savanna is a sandy, arid region in the western interior, with the project area falling across two eco-regions, namely the Kalahari Plain Thorn Bushveld (Olifantshoek Plains Thornveld, classified as Least Threatened) and the Kalahari Mountain Bushveld (Kuruman Mountain Bushveld, also classified as Least Threatened). Dwarf shrubs, shrubs and trees dominate the physiognomy, but open grassland plains are also present *[6]*.

Current and historical land-use:

The land on which the Redstone CSP plant is located was greenfield veld (bush) type, zoned as agricultural land. It was used for grazing horses, cattle and game. The current landowner also mines red jasper on a small scale on the north-western boundary of the site. The R385 road runs along the northern boundary of the project site and a railway line runs along the southwest of the project site. The Jasper and Lesedi solar PV facilities border the CSP plant on the eastern and southern boundaries respectively. The surrounding land uses also include agricultural activities with the mining village of Owendale located 2.5 km from the eastern boundary of the farm portion.

The area to the east of Postmasburg is predominantly agricultural, interspersed with mining activities. Other significant land uses in the area are:

- Various farmhouses and farm labourer residences.
- The residences in Groenwater Village (Metsimetala) and the village to the west of the Groenwater Siding.
- The Owendale residential township.
- The Lime Acres Mine residential township.
- The Goedgedacht/Jenn-Haven residential township.
- There are three schools in the vicinity:

    - Refentse Primary school in Groenwater Village.
    - Two schools in Lime Acres Mine Village.
- Recreationally facilities at the mine at Lime Acres and at Owendale.

Present and prior environmental conditions of the project area

Present and prior environmental conditions are as follows:

Climate

The area is also known as the Green Kalahari, with a hot and dry climate. During the summer months the temperatures can reach up to a maximum of 42°C in January. Rainfall patterns reveal that during an average year about 330 $mm.a^{-1}$ precipitation can be expected, however during wetter years rainfall of over 600 mm has been recorded, whilst during exceptionally dry years the annual precipitation merely reached 200 mm. Most of the rainfall received in the area is of convective origin and occurs in summer. Storms are relatively brief, but peak rainfall intensities over 5, 10 and 15 minutes differ little from other parts of South Africa which receive greater annual rainfall. About 5 days a month in January daily maximum temperatures reach over 35°C, with minimum relative humidities at midday dropping to 20%-30% and even lower. Frost can be severe in the winter months *[6]*.

Hydrology

Pan evaporation in the area is estimated at between 2,200 and 2,600 $mm.a^{-1}$, considerably more than the rainfall. The region is therefore in permanent water deficit and standing surface water, if it does not infiltrate, soon evaporates. Regional runoff can range from 0 - 25 $mm.a^{-1}$. Surface water generated by rainfall is confined to intense convective storms and quickly subsides. Streams in the vicinity are ephemeral, with the Groenwater Spruit running intermittently along the western border of the site. Annual recharge to groundwater in the area is about 3-10 $mm.a^{-1}$ *[6]*.

Topography

The topography of the area is hilly terrain, with low hills to the north, north-east and east of the site. The site on which the CSP plant is located is relatively flat. To the west, the ground rises slightly more steeply with a slope of about 1:50 or 2-3%. The ground is stony and has a large floating rock component (boulders not attached to the parent rock system).

Soils

The area is covered with sands that have a high porosity and infiltration capacity and can be classified as soils with a deep Hutton profile.

Vegetation

The vegetation is characterised by open shrubveld with *Lebeckia macrantha* prominent in places and a well-developed grass layer in place. The conservation status of these eco regions are set at Least Threatened, but none of this vegetation type is formally conserved in statutory conservation areas. Prior to the construction of the CSP facility, thetheThethe transformation status waswasiswas low, withwith some parts heavily utilised for grazing purposes.

The limited rainfall in this biome, coupled with fires and grazing pressures, prevent the upper layer from developing and keeps the grass layer dominant. While the savanna physiognomy of the area is indicated by the presence of several woody species comprising a relatively large proportion of diversity and dominance in some areas. During the environmental impact assessment phase the following vegetation species were observed in the project area: Grasses (37 species, 25.7%), forbs (46 species, 31.9%), shrubs (24 species, 16.7%), geophytes (14 species, 9.7%) and succulents (14 species, 9.7%). A total of 45 plant families are represented by the floristic diversity of the site, dominated by Poaceae (37, species, 25.7%) and Asteraceae (20 species, 13.9%) *[6]*.

1.15 Compliance with Laws, Statutes and Other Regulatory Frameworks

Compliance with provincial, national and international legislation was considered during the planning and implementation of this project. The following legislative aspects were taken into account:

Table 4: Relevant Legislation

Constitution of the Republic of South Africa (Act 108 of 1996)	The Bill of Rights, in the Constitution of South Africa (No. 108 of 1996), states that everyone has a right to a non-threatening environment and requires that reasonable measures are applied to protect the environment. This protection encompasses preventing pollution and promoting conservation and environmentally sustainable development. These principles are embraced in NEMA and given further expression.
National Development Plan, Agenda 2030,	The NDP envisages that, by 2030, South Africa will have an energy sector that provides reliable and efficient energy service at competitive rates, that is socially

	equitable through expanded access to energy at affordable tariffs and that is environmentally sustainable through reduced pollution.
Electricity Regulation Act (Act 4 of 2006) and the **Electricity Regulation on New Generation Capacity R399 of 4 May 2011, as amended by R1366 of 04 November 2016**	The act establishes a national regulatory framework for the electricity supply industry. It designates the National Energy Regulator of South Africa (NERSA) as the custodian and enforcer of this framework. The act covers licensing, registration and regulation of electricity generation, transmission, distribution, trading, import and export. The Regulations provide for the procurement of new generation capacity derived by organs of state, from renewable energy sources and co-generation; base load, mid-merit load and peak load new generation capacity; and cross border projects.
Integrated Resource Plan (IRP2019), and the draft IRP2023	The IRP is the electricity infrastructure development plan, of the Department of Energy, based on the least-cost electricity supply and demand balance, taking into account security of supply and the environment (minimize negative emissions and water usage).
National Environmental Management Act, (NEMA) (Act No. 107 of 1998)	Requires adherence to the principles of Integrated Environmental Management (IEA) in order to ensure sustainable development, which, in turn, aims to ensure that environmental consequences of development proposals be understood and adequately considered during all stages of the project cycle and that negative aspects be resolved or mitigated, and positive aspects enhanced.
National Environmental Management Biodiversity Act, (NEMBA). (Act No. 107 of 1998)	Provides for the management and conservation of South Africa's biodiversity within the framework of the National

	Environmental Management Act 1998 (NEMA); the protection of species and ecosystems that warrant national protection; the sustainable use of indigenous biological resources; the fair and equitable sharing of benefits arising from bioprospecting involving indigenous biological resources; the establishment and functions of a South African National Biodiversity Institute; and for matters connected therewith. The conservation of soil, water resources and vegetation are promoted. Management plans to eradicate weeds and invader plants must be established to benefit the integrity of indigenous life.
Northern Cape Nature Conservation Act. (Act No. 9 of 2009)	Provides for the protection and sustainable utilisation of plants, wild animals and aquatic biota; issuing of permits and other authorisations; and provides for offences and penalties for contravention of the Act.
Conservation of Agricultural Resources (Act 43 of 1983)	Provides for control over the utilization of the natural agricultural resources in order to promote the conservation of the soil, water sources and vegetation and the combating of weeds and invader plants; and for matters connected therewith.
National Water Act (No. 36 of 1998)	Provides the legal framework for the effective and sustainable management of water resources.
Occupational Health and Safety Act (No. 85 of 1993)	The Occupational Health and Safety Act, 1993, requires the employer to bring about and maintain, as far as reasonably practicable, a work environment that is safe and without risk to the health of the workers. This means that the employer must ensure that the workplace is free of hazardous substances, such as benzene, chlorine and micro-organisms, articles, equipment,

	processes, etc. that may cause injury, damage or disease. Where this is not possible, the employer must inform workers of these dangers, how they may be prevented, and how to work safely, and provide other protective measures for a safe workplace.
The Basic Conditions of Employment Act (No. 75 of 1997)	The act intends to give effect to the right to fair labour practices referred to in section 23(1) of the Constitution by establishing and making provision for the regulation of basic conditions of employment; and thereby to comply with the obligations of the Republic as a member state of the International Labow Organisation; and to provide for matters connected therewith.
Carbon Tax Act 15 of 2019 and supporting regulations,	Provide for VCUs to be used to offset carbon tax liability.

1.16 Double Counting and Participation under Other GHG Programs

1.16.1 No Double Issuance

Is the project receiving or seeking credit for reductions and removals from a project activity under another GHG program?

☐ Yes ☒ No

1.16.2 Registration in Other GHG Programs

Has the project registered under any other GHG programs?

☐ Yes ☒ No

Is the project active under the other program?

☐ Yes ☒ No

1.16.3 Projects Rejected by Other GHG Programs

Has the project been rejected by any other GHG programs?

☐ Yes ☒ No

1.17 Double Claiming, Other Forms of Credit, and Scope 3 Emissions

1.17.1 No Double Claiming with Emissions Trading Programs or Binding Emission Limits

Are project reductions and removals or project activities also included in an emissions trading program or binding emission limit? See the *VCS Program Definitions* for definitions of emissions trading program and binding emission limit.

☐ Yes ☒ No

1.17.2 No Double Claiming with Other Forms of Environmental Credit

Has the project activity sought, received, or is planning to receive credit from another GHG-related environmental credit system? See the *VCS Program Definitions* for definition of GHG-related environmental credit system.

☐ Yes ☒ No

1.17.3 Supply Chain (Scope 3) Emissions

Do the project activities specified in Section 1.12 affect the emissions footprint of any product(s) (goods or services) that are part of a supply chain?

☐ Yes ☒ No

1.18 Sustainable Development Contributions

1.18.1 Sustainable Development Contributions Activity Description

The sustainability objective of the project lies in its capacity to reduce greenhouse gas emissions. In alignment Over the seven-year crediting period, renewable twice, the project anticipates an average annual emission reduction of 500,000 tons of CO2 equivalent, totalling 10,500,000 tons of CO2 equivalent over the project lifetime. This substantial reduction aligns with Sustainable Development Goal (SDG) 13 (Climate Action), specifically addressing the urgent need to combat climate change and its impacts. Monitoring will be undertaken as per the plan in this Project Description, which involves the monitoring of generated renewable energy in MWh/year. This aligned with South African SDG 7.2 which aims to increase the renewable energy within the country. The project displaces approximately 480 000 - 500 000 MWh per annum of emission intensive grid electricity through the generation of low emission solar CSP electricity.

Furthermore, aligned with SDG 8.5, the project will promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all. This will be achieved through just less than 500 jobs being created during the construction phase and around 40 jobs during the ongoing operation of the project into the future.

1.18.2 Sustainable Development Contributions Activity Monitoring

The project activities that result in Sustainable Development (SD) contributions are summarised in the table below:

Row number	SDG Target	SDG Indicator	Net Impact on SDG Indicator	Current Project Contributions	Contributions Over Project Lifetime
1)	7.2	7,2,1A1 (South African SDG Indicators) – Amount of renewable energy at annual operating capacity	Implemented activities will increase the amount of renewable energy in South Africa	Displaces approximately 480 000 - 500 000 MWh per annum of emission intensive grid electricity through the generation of low emission solar CSP electricity.	Displaces approximately 10 500 GWh of emission intensive grid electricity through the generation of low emission solar CSP electricity.

Row number	SDG Target	SDG Indicator	Net Impact on SDG Indicator	Current Project Contributions	Contributions Over Project Lifetime
2)	13.0	Tonnes of greenhouse gas emissions avoided or removed	The total GHG emissions of the country will be reduced by the ERs and REMs that will no longer be contributing to the country's total GHG emissions. It is expected that the number of GHG emissions will be reduced annually as the project will be verified, confirming that these are no longer emitted to the atmosphere.	Displaces approximately 500,000 tCO_2e per annum of emission intensive grid electricity through the generation of low emission solar CSP electricity.	Displaces approximately 10,500,000 tCO_2e of emission intensive grid electricity through the generation of low emission solar CSP electricity over the 21-year project lifetime.

Row number	SDG Target	SDG Indicator	Net Impact on SDG Indicator	Current Project Contributions	Contributions Over Project Lifetime
3)	8.5	Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all	With total unemployment in South Africa having risen from 24.9% to 33.5% over the past decade (2012 to 2022) this SDG target remains a high priority for the country. The country has a target to, "By 2030, achieve full and productive employment and decent work for all women and men, including for young people and persons with disabilities, and equal pay for work of equal value."	During the construction phase there will be 481 jobs created and during Operation there will be 41 jobs. [6]	The project planned to create 481 jobs during the construction phase (from domestic labour)[4] and 41 jobs during operations[5] [6]. The Operational jobs will be ongoing, but the construction jobs would only be during the construction phase and will phase out over a period once the plant is operational.

[4] Table 3-3 of Socio-economic assessment of the project

[5] Table 3-4 of Socio-economic assessment of the project

1.19 Additional Information Relevant to the Project

1.19.1 Leakage Management

As per the applied methodology ACM0002 (Ver. 22.0), no leakage emissions are considered in the project activity.

1.19.2 Commercially Sensitive Information

No commercially sensitive information has been excluded from the public version of the project description.

1.19.3 Further Information

No further information.

2 SAFEGUARDS AND STAKEHOLDER ENGAGEMENT

2.1 Stakeholder Engagement and Consultation

2.1.1 Stakeholder Identification

Stakeholder Identification	An extensive stakeholder engagement process was conducted as part of the environmental impact assessment that took place during the scoping phase of the project. The following measures were implemented to identify stakeholders and to request members of the public to register as interested and affected parties (I&APs): • The project was advertised in two local newspapers (in Afrikaans, English and Setswana). • A background information document was distributed to stakeholders via email and placed in three local public libraries. • Three site notices were placed on the northern boundary of the site. • Pamphlets and notices were printed on A5 papers and distributed via the South African post offices to +- 1900 post boxes in Postmasburg and Daniëlskuil. • A4 and A3 site notices were place on notice boards at eight public amenities in Postmasburg and Daniëlskuil. All interested and affected party (I&AP) details were compiled into a database which includes contact details, dates and details of consultations, and a record of all issues raised, and responses provided. This database will be updated on an on-going basis. Key stakeholders include representatives from: • Department of Forestry, Fisheries and the Environment (DFFE)

	• Department of Water and Sanitation (DWS) • South African Heritage Resource Agency (SAHRA) • Relevant Northern Cape provincial authorities (environment, conservation, agriculture) • Relevant local and district municipalities • Parastatals including Eskom and the local aviation authority • Air traffic and navigation systems • Affected and surrounding landowners • Environmental non-governmental organisations (wildlife society of South Africa, BirdlifeSA) • Community based organisations • Residents of surrounding communities and towns • Surrounding businesses, power plants and mines
Legal or customary tenure/access rights	The legal tenure of the land is private ownership, and a binding lease agreement is in place with the landowner, in order to establish the Redstone CSP facility on his farm land.
Stakeholder diversity and changes over time	Key stakeholders include government agencies, neighbouring landowners and residents, local communities, businesses and mines. The mining sector is the key role player in the economies surrounding the project area, followed by electricity generation and construction. As such, these sectors play an important role in the social, cultural and economic fabric of the surrounding communities. Because these sectors are strictly regulated, they provide mostly permanent employment and as such, casual employment is limited. Approximately 45 – 50% of the surrounding communities live in poverty [7]. Changes in the make-up of the group are anticipated to be minimal throughout the duration of the project.

Expected changes in well-being	No significant negative impacts on the well-being of stakeholders are anticipated to occur because of this project. There are many positive impacts, as documented in the project's economic development plan [7]. This includes, among others: • Increased production / business sales • Increase in local Gross Domestic Product (GDP) • Increase in employment opportunities • Increase in household income • Improved housing and basic service provision.
Location of stakeholders	The stakeholders that are most likely to be impacted by the project reside in the surrounding towns and communities of: • Postmasburg • Daniëlskuil • Groenwater • Lime Acres • Owendale
Location of resources	The Redstone CSP facility is located on the remainder of the Farm 469, the Hay District, Tsantsabane Local Municipality of the ZF Mcgawu District, Northern Cape Province, South Africa. The landowner has granted the owner of the CSP plant the rights to lease the land, and will therefore not himself have grazing access to the area for the next 25 years.

2.1.2 Stakeholder Consultation and Ongoing Communication

Date of stakeholder consultation	August – December 2011 and August-2024
Stakeholder engagement process	An extensive stakeholder engagement process formed part of the Environmental Impact Assessment (EIA) and was carried out during the project design phase in August 2011 and communication has been upheld since then [6]. This included stakeholder identification as set out in 2.1.1, followed by two public meetings, held at: • Postmasburg Town Hall on 25 August 2011 at 17:30 • Groenwater Community Hall on 26 and 29 August 2011 at 17:30 The purpose of the public meetings was to provide information about the project and to provide stakeholders with a platform to provide input to the project planning process. The meetings were hosted in a culturally appropriate and inclusive manner, and communication was facilitated in the three locally prominent official languages (Afrikaans, English, Setswana). As a result of the great effort invested in hosting the initial public meetings and the poor attendance by the greater part of the affected communities invited to those meetings, the DFFE was consulted to request an exemption from the requirements of hosting public meetings for the remainder of the public consultation process. The DFFE approved the exemption and subsequently all engagements with stakeholders and I&APs have been via written and/or electronic correspondence. Given the long timeline for the implementation of this project, communication with the registered I&APs has been ongoing, and the database has been expanded and kept up to date. The additional stakeholder consultation that was carried out specifically in relation to the VCS requirements, built on this existing communication trail. All stakeholders in the registered I&AP database were contacted directly via phone calls and/or emails, and notices were printed out and put up at the following public locations in:

	Daniëlskuil: • Friendly Grocer • OK Foods • Kgatelopele Local Municipality • Siyanda District Offices of the Department of Social Services and Population Development. Lime Acres: • Lime Acres Family Store Owendale village: • Entrance gate to the Owendale village Postmasburg: • Spar • Saverite Photographs of the notices displayed in public locations are shown in Figure 7 and Figure 8.
Consultation outcome	All issues raised during the public consultation process have been included in the issues and responses report which forms part of the EIA [6]. All issues that were recorded have been resolved. In summary, no project or environmental related concerns have been raised by registered I&APs nor have any objections been received in this regard. Additionally, the project has been received very well by all relevant commenting authorities and no objections or opposition has been received. To date, no issues or comments have been received in relation to the VCS validation and verification process.
Ongoing communication	Ongoing communication will be via direct electronic communication to stakeholders recorded in the registered I&AP database, as well as public notices placed in public locations in the surrounding communities. In addition, grievance boxes have been permanently installed at selected public location, and example is

	shown in Figure 9. These are monitored on a continual basis and any grievances received will be recorded and addressed.
Stakeholder input	All comments received during the EIA and stakeholder consultation has been carefully considered and addressed in the EIA and project implementation. No updates to the project design has been considered necessary. Stakeholder comments will continue to be monitored to consider whether further updates to the project are required.

Figure 7: Notices at the Daniëlskuil OK Grocer

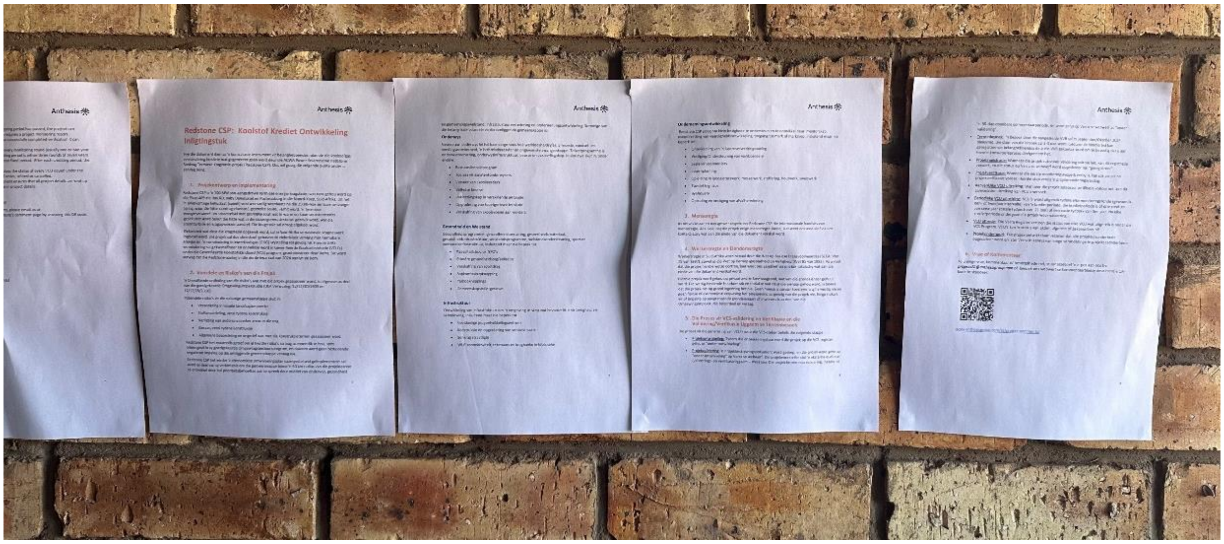

Figure 8: Notices at the Postmasburg Community Hall

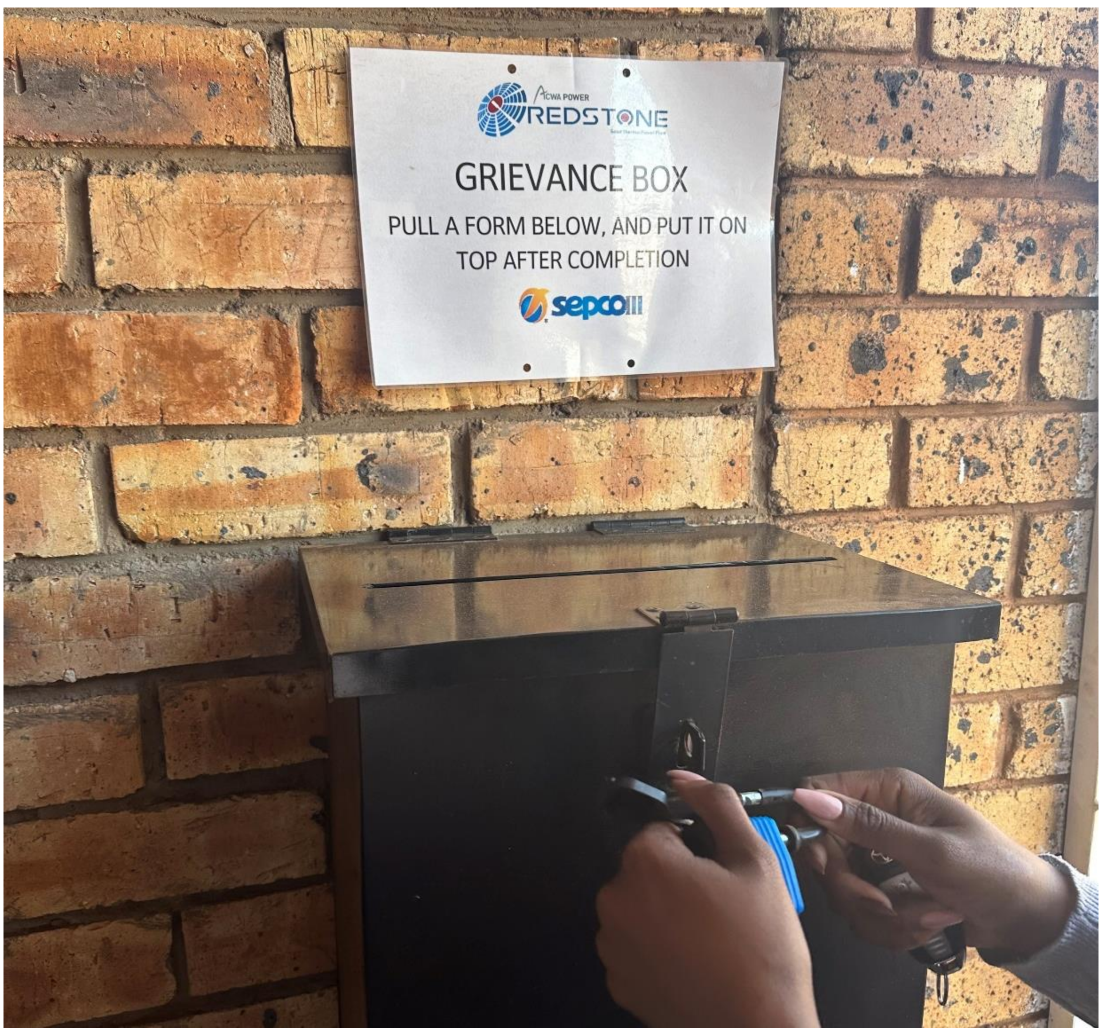

Figure 9: Permanently Installed Grievance Box at Daniëlskuil community hall.

2.1.3 Free Prior and Informed Consent

Obtaining consent	The landowner has consented to the project by granting a lease agreement to install the CSP infrastructure on his property. No IPs (Indigenous Peoples), LCs (Local Communities) or customary rights holders are impacted by the Redstone CSP project activities.
Outcome of FPIC	The outcome of the negotiations is that the Redstone CSP facility will be established on the private land for the next 25 years, along with the Lesedi and Jasper PV facilities which both have lease agreements in place with the landowner since 2011.

2.1.4 Grievance Redress Procedure

Development process	Grievances can be received via three channels: • The grievance portal on the Anthesis website • Grievance boxes permanently situated at various public locations in the project's vicinity and at the Redstone site security. • There are also permanently employed community liaison officers stationed in the surrounding communities with the goal of keeping communication up to date with development at Redstone. Any grievances received will be handled on a case-by-case basis by Anthesis in consultation with the Redstone project team.
Grievance redress procedure	Any grievances, complaints or conflicts from stakeholders would be captured via a dedicated feedback form on the Anthesis website, or from the other channels mentioned above. The Grievance Redress process consists of: • Receiving the issue from a stakeholder • Assigning it to the relevant manager/team

	• Responding to the stakeholder (which may include telephonic, written or face to face communication) in an attempt to understand and resolve the issue in good faith • Recording the outcome of the interaction. • Grievances that cannot be resolved amicably in this manner will be referred to mediation by a neutral third party and failing resolution shall be raised to either arbitration or courts of law in South Africa. Stakeholders have an ongoing opportunity to log into a dedicated feedback form on the Anthesis website in order to leave comments and engage with the project team. This website will be part of the ongoing and continuous engagement between the project and stakeholders. Comments received will be logged and taken into account in project operation and future verifications.

2.1.5 Public Comments

Comments received	Actions taken
This section will be completed once the public commenting period has passed. No comments received at time of publishing.	This section will be completed once the public commenting period has passed

2.2 Risks to Stakeholders and the Environment

2.2.1 Management Experience

The Anthesis SA team has significant experience in implementing similar project activities, engaging with communities, project management and environmental and social safeguards aligned with the VCS requirements. Members of the project development team have in excess of 50 collective years' experience in the voluntary carbon market. Members of the project development team have more than 50 collective years' experience in the voluntary carbon market. They have been originating, developing and monetizing carbon credits as a company for over 2 decades. The Redstone project proponent has extensive renewable energy project experience in the greater Middle East and Africa region, including similar Concentrated Solar Power facilities at Bokpoort, South Africa; Noor Energy, United Arab Emirates and NOOR I & II, Morocco.

The integrated team is well positioned to implement and operate the facilitate and deliver the associated GHG emission reductions into the carbon market.

2.2.2 Risk Assessment

Potential risks to stakeholders and the environment, as well as mitigation measures, are described in the table below:

	Risks identified	Mitigation or preventative measure(s) taken
Natural and human-induced risks to stakeholders' wellbeing	Construction activities will cause minor change in landscape characteristics over the localized area effecting the visual quality of the area.	The heliostat terrace is established at the lowest level possible to take advantage of the surrounding topography which acts as an effective visual screen (especially to views from the south and west of the site). An ecological approach to rehabilitation measures, as opposed a horticultural approach will be adopted wherever possible. Communities of indigenous, preferable endemic, plants will be established in order to enhance biodiversity and blend with existing vegetation. A registered landscape architect (SACLAP) will be consulted for this purpose. The tower should remain as a mainly concrete finish and no advertising should be allowed on it.
	Impacts on air quality caused by dust	During the construction phase, measures will be implemented to reduce the amount of dust in the air resultant from construction activities.

		During the operational phase the following mitigation measures will be applied: vegetation levels below the heliostats will be maintained to ensure no exposed surfaces are present for the liberation of dust from within and surrounding the site. Larger trees will be planted surrounding the site to act as wind breaks and reduce the wind speeds within the plant area. A perimeter fence that reduces wind on the heliostats will mitigate dust formation.
	Light pollution	Light pollution will be kept to a minimum by implementing the following measures: Security and flood lighting will only be used where absolutely necessary and carefully directed i.e. away from nearby residences, roads and communities. Wherever possible, lights will be directed downwards to avoid illuminating the sky. Heliostats will be maintained to prevent reflecting into adjacent land areas. Use of security lighting at the periphery of the site that is activated by movement and are not permanently kept on. Install light fixtures that provide precisely directed illumination to reduce light "spillage" beyond the immediate surrounds of the Project Site.

	Noise pollution during construction	The following noise level mitigation measures will be implemented during construction phase: Use of low-noise generation construction machinery, curtailing the uses of reverse-warning signals on site vehicles in certain areas and at certain times (i.e. outside normal working hours), truck traffic should be routed away from noise sensitive areas, etc. During operation, the design of all major equipment for the plant is to incorporate all the necessary acoustic design aspects required so that the overall generated noise level from the new installation does not exceed a maximum equivalent continuous day/night rating level of 70dBA (just inside the property projection plane, namely the property boundary of the CSP Plant) as specified for industrial districts in SANS 10103.
	Visual impact of general construction activities, the potential scarring of the landscape due to vegetation clearing.	The following extract from the approved environmental management programme (EMPr) applies: Keep vegetation removal to a minimum where possible. Ensure, wherever possible, all existing vegetation is retained and incorporated into the site rehabilitation. Ensure that all infrastructure and the site and general surrounds are maintained in a neat manner.

Risks to stakeholder participation	No risk identified	Stakeholders have communicated with the project developer electronically and in writing over the past years during project development phases. No project or environmental related concerns have been raised by registered stakeholders nor had any objections been received in this regard. Additionally, the project has been received very well by all relevant government authorities and no objections or opposition has been received.
Working conditions	No Risk identified	The Operating proponent ensures that working conditions comply with local legislation, specifically the Occupational Health and Safety Act (Act No 85 of 1993) and Basic Conditions of Employment Act (No. 75 of 1997).
Safety of women and girls	No Risk identified	Because the project site is not open to the public, no risks to women and girls were identified on the project site, furthermore Redstone's employment policies adhere to the Occupational Health and Safety Act and the Basic Employment Conditions Act which protect the rights of both women and men in the workplace and therefore no risks were identified. In addition, Redstone's Code of Conduct and Ethics Policy specifically addresses a range of topics including non-harassment and non-discrimination [8].
Safety of minority and marginalized groups, including children	No risk identified	Because the project site is not open to the public, no risks to minority and marginalized groups, including children were identified

		on the project site. Section 28 of the Constitution of South Africa specifically protects children from exploitative labour practices, from work that is inappropriate for a child's age, and work that puts the child's education and physical, mental, spiritual, moral and social development at risk. Since Redstone's Employment and Code of Conduct and Ethics Policies adhere to these laws, no risks were identified [8].
Pollutants (air, noise, discharges to water, generation of waste, and release of hazardous materials and chemical pesticides and fertilizers)	Brine from concentrating ponds leaching into the groundwater	The industrial wastewater to be generated by the project is classified as a moderate hazard, with a Hazard rating of 3. The brine is an inorganic process waste or residue and was classified as class 6 (Poisonous (toxic) substances) according to the SABS 0228 code. The ponds will therefore be lined with a triple liner and double drainage system as required by the Department of Water and Sanitation (DWS) [9].

2.3 Respect for Human Rights and Equity

2.3.1 Labor and Work

The table below provides a summary of the identified risks related to labor and work, as well as the measures in place to prevent and mitigate these risks.

	Risks identified[6]	Mitigation or preventative measure(s) taken
Discrimination	No risk identified	Risk is mitigated as per Redstone's code of conduct and ethics policy [8]: Redstone has

[6] The identified risks and commensurate mitigation or preventative measure(s) for forced labor, child labor, and human trafficking, must be inclusive of staff and contracted workers employed by third parties.

		established clear policies and guidelines to prevent discrimination, promote diversity and inclusion, and maintain a respectful and inclusive work environment for all employees, including contracted workers employed by third parties. The measures include promoting diversity and inclusion, non-harassment and non-discrimination policies, preventing corruption, preventing money laundering and financing of terrorism, government relations compliance, whistleblowing encouragement, open door policy, and reporting channels for addressing concerns without fear of discrimination or reprisal.
Sexual harassment	No risk identified	Risk is mitigated as per Redstone's code of conduct and ethics policy [8]: Redstone has implemented specific measures to prevent sexual harassment in the workplace, including a Non-Harassment & Non-Discrimination Policy, respectful behavior expectations, reporting channels, compliance training, and an open-door policy. Redstone Power aims to create a safe and respectful work environment free from sexual harassment by implementing these measures
Equal pay for equal work	No risk identified	Risk is mitigated as per Redstone 's code of conduct and ethics policy [8]: Redstone ensures equal pay for equal work by maintaining a fair and non-discriminatory compensation structure. The company values equality and fairness in remuneration, ensuring that employees receive equitable pay for the same or similar work regardless of personal characteristics such as gender, age, race, or religion. This commitment to equal pay for equal work aligns with Redstone 's principles of integrity, professionalism, and respect for all employees.
Gender equity in labor and work	No risk identified	Risk is mitigated as per Redstone 's code of conduct and ethics policy [8]: Redstone promotes gender equity by fostering a work

		environment that values diversity and inclusion, ensuring equal opportunities for career advancement, training, and development for all employees regardless of gender. The company's commitment to gender equity is reflected in its non-discrimination policies and initiatives aimed at creating a workplace free from bias and discrimination.
Forced labor	No risk identified	Risk is mitigated as per Redstone 's code of conduct and ethics policy [8]: Redstone strictly prohibits any form of forced labor, modern slavery, or human trafficking within its operations and supply chain. The company upholds international labor standards and human rights principles to ensure all employees, contractors, and suppliers are treated with dignity and respect. Redstone has a zero-tolerance policy towards forced labor and slavery, aligning with its commitment to ethical business practices and corporate social responsibility.
Child labor	No risk identified	Risk is mitigated as per Redstone 's code of conduct and ethics policy [8]: Redstone is committed to preventing child labor and exploitation in its operations and supply chain. The company ensures that all employees are of legal working age and complies with local laws regarding youth employment. Redstone condemns the use of child labor in any form and adheres to international conventions and laws prohibiting child labor. The company expects its business partners and associates to comply with relevant laws to prevent child labor. Redstone ‘s policies emphasize ethical sourcing and responsible business conduct to prevent and eliminate child labor practices.
Human trafficking	No risk identified	Risk is mitigated as per Redstone 's code of conduct and ethics policy: Redstone strictly prohibits any form of forced labor, modern slavery, or human trafficking within its operations and supply chain. The company

		upholds international labor standards and human rights principles to ensure all employees, contractors, and suppliers are treated with dignity and respect. Redstone has a zero-tolerance policy towards forced labor and slavery, aligning with its commitment to ethical business practices and corporate social responsibility

2.3.2 Human Rights

The table below provides a summary of risks identified related to human rights in line with applicable international human rights law, and the United Nations Declaration on the Rights of Indigenous Peoples and ILO Convention 169 on Indigenous and Tribal Peoples.

Risks identified	Mitigation or preventative measure(s) taken
No risk identified	The project company adheres to international and national human right laws. The EIA undertaken as part of the project included an in-depth socio-economic impact assessment, which included impacts on surrounding indigenous communities [6]. Negative impacts were considered negligible and mitigated as far as possible and are by far outweighed by the positive impacts this project will have on neighbouring communities.

2.3.3 Indigenous Peoples and Cultural Heritage

A heritage and archaeological impact assessment was carried out as part of the EIA. The assessment has shown that the area between Postmasburg and Daniëlskuil, generally referred to as the Ghaap plato has a rich history of occupation from the Stone Age with hunter gatherers to the Thlaping and Thlaro during the Iron Age period. The 1800's saw the rise of the Griqua people in the area and their loss of sovereignty after 1880 to Cape rule. The field work identified a total of 25 heritage sites. The overall impact of the development on heritage resources was found to be acceptably low and impacts can be mitigated to acceptable levels. The Environmental Authorization for this development was granted by the authorities after due consideration of these impacts.

Risks identified	Mitigation(s) or preventative measure taken
Destruction of archaeological sites because of construction activities	The following extract from the approved environmental management programme (EMPr) applies: If any evidence of archaeological sites or remains (e.g. remnants of stone-made structures, indigenous ceramics, bones, stone artefacts, ostrich eggshell fragments, charcoal and ash concentrations), fossils or other categories of heritage resources are found during the proposed development, work in that area must be stopped immediately. The SAHRA APM Unit must be alerted. If unmarked human burials are uncovered, the SAHRA Burial Grounds and Graves (BGG) Unit, must be alerted immediately. A professional, depending on the nature of the finds, must be contracted as soon as possible to inspect the findings.
Destruction of cemeteries or graves because of construction activities.	The site layout has been adjusted to avoid cemeteries and the cemeteries are fenced with a 10-meter buffer.

2.3.4 Property Rights

The table below provides a summary of risks related to property to protecting and preserving the property rights of IPs (Indigenous Peoples), LCs (Local Communities) and customary rights holders, and to protecting legal or customary tenure/access rights to territories, property, and resources

Risks identified	Mitigation or preventative measure(s) taken
No risks identified	The project is implemented on privately owned agricultural land for which legal land tenure has been secured via a lease agreement is in place. As such, no risks to property rights were identified, and no encroachment on community land is taking place.

2.3.5 Benefit Sharing

The project does not impact on property rights and therefore no benefit sharing agreement is required. However, the project goes to significant lengths to uplift the surrounding communities by implementing its economic development plan [7]. This will be implemented as part of its commitment to uplift and develop communities within a 50 km radius of the project site by addressing their priority needs through education, health and community wellness, infrastructure improvement and enterprise development. Some of the main benefits to the surrounding communities are:

Education

Lack of education results in high unemployment rate, poverty, food and nutrition insecurity, a high crime rate and unwanted pregnancies. Target programmes are education equipment, educational infrastructure, bursaries and apprenticeship. Initiatives include, among others:

- Drivers' license program
- Stipends for assistant teachers
- Transportation of school learners
- Full time bursaries
- Apprenticeship in different trades
- Upgrades to toilet facilities at schools
- Supply of school shoes to learners

Health and wellness

Health programmes, health equipment, health materials, health infrastructure, wellness programmes, wellness support, sports, and wellness infrastructure including but not limited to:

- Capacity building for non-profit organizations (NPOs)
- Primary healthcare facilities
- Supply of sports apparel
- Soup kitchen renovation
- Park renovations
- Community toilet facilities

Infrastructure

Development of infrastructure to create an enabling environment for business and society including but not limited to:

- Multipurpose youth development center
- Construction and upgrading of various roads
- Solar streetlights
- Wi-fi connectivity, desktops and power banks in library

Enterprise development

Redstone CSP aims to support and development Small and Medium Enterprises (SMMEs) through an incubator, mentorship, value chain of skills development, access to market or market creation, including but not limited to:

- Development of a lucerne processing plant
- Establishment/Support of pig farming
- Scales for livestock farming
- Lucerne planting
- Training in plumbing, bricklaying, upholstery, carpentry, welding
- Appliance technician
- Honeybee keeping
- Training and establishment of waste recycling

2.4 Ecosystem Health

The table below provides a summary of risks related to impacts on biodiversity and ecosystems, soil degradation and soil erosion, and water consumption and stress

	Risks identified	Mitigation or preventative measure(s) taken
Impacts on biodiversity and ecosystems	Damage to indigenous natural vegetation and sensitive areas on development footprint	As per the approved EMPr: All disturbed areas within the Project Development Footprint will be rehabilitated as soon as construction is complete within an

		area. The terms of the EMPr and the Environmental Authorisation were included in all tender documentation and Contractors' contracts. The extent of clearing and disturbance to the native vegetation is kept to a minimum so that impact on flora and fauna is restricted.
	Impacts on avifauna in project area	As per EMPr: A faunal register must be maintained to record avifaunal injuries and fatalities. The environmental control officer (ECO) must be trained by an avifaunal specialist to identify potential Red Data species as well as the signs that indicate possible breeding by these species. The ECO must then, during audits/site visits, make a concerted effort to look out for such breeding activities of Red Data species, and such efforts may include the training of construction staff (e.g. in Toolbox talks) to identify Red Data species, followed by regular questioning of staff as to the regular whereabouts on site of these species.
Soil degradation and soil erosion	No risk identified	No risks were identified on soil in the operation of the plant. Construction activities were mitigated as per the EMPr.
Water consumption and stress	No risk identified	There will be no risk or impacts on water consumption and stress in the operation of the plant which is located on private property. Construction activities were mitigated as per the EMPr.

2.4.1 Rare, Threatened, and Endangered Species

Is the project located in or adjacent to habitats for rare, threatened, or endangered species?

☒ Yes ☐ No

<u>Fauna:</u>

A total of 56 Red Data animals (excluding avifauna) are known to occur in the Northern Cape Province. Of these species, 12 are listed as Data Deficient (DD), 21 are listed as Near Threatened (NT), 12 are listed as Vulnerable (VU), 5 are listed as Endangered (EN) and 5 species are listed as Critically Endangered (CR). It is estimated that approximately 79% of these species have a low probability of occurring in the immediate region of the project area. 11 species are estimated to have a moderate probability of occurring and two species have a high probability of occurrence in the immediate region of the project area. All species with moderate or higher probability of occurrence are shown in Table 5 below.

Table 5: Red Data Species in the Project Area

Biological Name	English Name	Status	Probability or occurrence
Pyxicephalus adspersus	Giant Bullfrog	Near Threatened	moderate
Atelerix frontalis	South African Hedgehog	Near Threatened	confirmed
Cistugo lesueuri	Leseur's Wing-gland Bat	Near Threatened	moderate
Crocidura cyanea	Reddish-grey Musk Shrew	Data Deficient	moderate
Hyaena brunnea	Brown Hyaena	Near Threatened	confirmed
Manis temminckii	Pangolin	Vulnerable	high
Mellivora capensis	Honey Badger	Near Threatened	moderate
Miniopterus schreibersii	Schreiber's Long-fingered Bat	Near Threatened	moderate
Paratomys littledalei	Littledale's Whistling Rat	Near Threatened	moderate
Poecilogale albinucha	African Weasel	Data Deficient	moderate
Rhinolophus clivosus	Geoffroy's Horseshoe Bat	Near Threatened	moderate
Rhinolophus darlingi	Darling's Horseshoe Bat	Near Threatened	moderate
Rhinolophus denti	Dent's Horseshoe Bat	Near Threatened	moderate
Suncus varilla	Lesser Dwarf Shrew	Data Deficient	confirmed
Tatera leucogaster	Bushveld Gerbil	Data Deficient	high

<u>Avifauna:</u>

South African Bird Atlas Project (SABAP) records 168 bird species of which 11 are Red Listed Species. The presence of red data species in the project area was assessed based on the avifaunal impact assessment specialist study carried out as part of the EIA process. The site does not fall within an Important Bird Area (IBA) and there were no IBA's within proximity to the site. The likelihood of occurrence of red data species in the project area is shown in Table 6 below:

Table 6: Red Data Avifauna in the Project Area

Species	Conservation status	Likelihood of occurrence
Martial Eagle	VU	Possible
Lesser Kestrel	VU	Likely
Blue Crane	VU	Likely
White-backed Vulture	VU	Possible
Secretary bird	NT	Possible
Greater Flamingo	NT	Possible

Species and habitat	The nature of the proposed development is expected to result in limited indirect impacts on the fauna species listed above. However, the loss of a small portion of this habitat type is not expected to result in significant impacts on a regional scale since much of this habitat are present to the north of the project site, while the regional vegetation type is also afforded a Least Threatened status (VEGMAP). The implementation of site specific and generic mitigation measures, as per the approved EMPr, together with development recommendations is expected to lower the expected impacts to an acceptable level.
Areas needed for habitat connectivity	The general region is characterised by extremely low levels of transformation and habitat fragmentation. Impacts from the proposed development are unlikely to increase regional or local levels of fragmentation and habitat isolation significantly.

The table below summarizes the identified risks related to habitats of rare, threatened, and endangered species, and of habitat connectivity, along with the mitigation measures in place to minimize the risks.

	Risks identified	Mitigation or preventative measure(s) taken
Habitats for rare, threatened, and endangered species	Potential impacts of the project on avifauna may include collision of birds	Fauna and avifauna will be checked periodically to ensure any undesirable impacts are mitigated. No specific guidance was given

	with heliostats and burning of birds in focal points.	during the EIA process, and this will be checked as required during operation.
	Loss of endangered fauna due to human interaction / during operational phases of the project activities.	As per EMPr: A faunal removal plan/ rescue plan will be implemented with designated competent personnel. Ensure the absence of larger animals through normal security and operational patrols. Ensure animal capture/ removal/ transportation equipment is available on site, such as snake hooks, tongs, bags, eye shield, etc.
Areas for habitat connectivity	No risk identified	The general region is characterised by extremely low levels of transformation and habitat fragmentation. Impacts from the proposed development are unlikely to increase regional or local levels of fragmentation and habitat isolation significantly.

2.4.2 Introduction of Species

No invasive species will be used as part of the project activities.

2.4.3 Ecosystem Conversion

This section is not applicable, seeing that this is not an ARR, ALM, WRC or ACoGS project. The Environmental Authorization was granted by the authorities to clear the land for establishing the CSP solar facility.

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

Type (methodology, tool or module).	Reference ID, if applicable	Title	Version
Methodology	ACM0002	Grid-connected electricity generation from renewable sources	22.0
Tool	TOOL01	Tool for the demonstration and assessment of additionality	7.0
Tool	TOOL07	Tool to calculate the emission factor for an electricity system	7.0
Tool	TOOL32	Positive lists of technologies	4.0

3.2 Applicability of Methodology

The project activity(s) meets each of the applicability conditions of the methodology(s), tools, and modules applied by the project (where applicable), as described below:

Methodology ID	Applicability condition	Justification of compliance
ACM0002	This methodology is applicable to grid-connected renewable energy power generation project activities that: (a) Install a Greenfield power plant; (b) Involve a capacity addition to (an) existing plant(s); (c) Involve a retrofit of (an) existing operating plant(s)/unit(s);	The project activity involves installation of a greenfield power plant (a) that is connected to the national grid of South Africa.

Methodology ID	Applicability condition	Justification of compliance
	(d) Involve a rehabilitation of (an) existing plant(s)/unit(s); or (e) Involve a replacement of (an) existing plant(s)/unit(s); or (f) Install a Greenfield power plant together with a grid-connected Greenfield pumped storage power plant. The greenfield power plant may be directly connected to the PSP or connected to the PSP through the grid.	
	In case the project activity involves the integration of a BESS, the methodology is applicable to grid-connected renewable energy power generation project activities that: (a) Integrate BESS with a Greenfield power plant; (b) Integrate a BESS together with implementing a capacity addition to (an) existing solar photovoltaic1 or wind power plant(s)/unit(s); (c) Integrate a BESS to (an) existing solar photovoltaic or wind power plant(s)/unit(s) without implementing any other changes to the existing plant(s); (d) Integrate a BESS together with implementing a retrofit of (an) existing solar photovoltaic or wind power plant(s)/unit(s); (e) Integrate a BESS together with a Greenfield power plant that is operating in coordination with a PSP. The BESS is located at site of the greenfield renewable power plant.	This project activity does not include the integration of a BESS

Methodology ID	Applicability condition	Justification of compliance
	The methodology is applicable under the following conditions:	
	(a) Hydro power plant/unit with or without reservoir, wind power plant/unit, geothermal power plant/unit, solar power plant/unit, wave power plant/unit or tidal power plant/unit;	(a) This project is a concentrated solar thermal power plant.
	(b) In the case of capacity additions, retrofits, rehabilitations or replacements (except for wind, solar, wave or tidal power capacity addition projects) the existing plant/unit must have started commercial operation prior to the start of a minimum historical reference period of five years. The reference period is used for the calculation of baseline emissions and defined in the baseline emission section. Furthermore, no capacity expansion, retrofit, or rehabilitation of the plant/unit has been undertaken between the start of this minimum historical reference period and the implementation of the project activity;	(b) This project is not a capacity addition, retrofit, rehabilitation or replacement, but a greenfield plant.
	(c) In case of Greenfield project activities applicable under paragraph 7(a) above, the project participants shall demonstrate that the BESS was an integral part of the design of the renewable energy project activity (e.g., by referring to feasibility studies or investment decision documents);	Not applicable to this project as a BESS is not installed.
	(d) The BESS should be charged with electricity generated from the associated renewable energy power plant(s). Only during exigencies2 may	Not applicable to this project

Methodology ID	Applicability condition	Justification of compliance
	the BESS be charged with electricity from the grid or a fossil fuel electricity generator. In such cases, the corresponding GHG emissions shall be accounted for as project emissions following the requirements under section 5.4.4 below. The charging using the grid or using fossil fuel electricity generator should not amount to more than 2 per cent of the electricity generated by the project renewable energy plant during a monitoring period. During the time periods (e.g., week(s), months(s)) when the BESS consumes more than 2 per cent of the electricity for charging, the project participant shall not be entitled to issuance of the certified emission reductions for the concerned periods of the monitoring period. (e) In case the project activity involves PSP, the PSP shall utilize the electricity generated from the renewable energy power plant(s) that is operating in coordination with the PSP during pumping mode.	Not applicable to this project
	In case of hydro power plants, one of the following conditions shall apply: (a) The project activity is implemented in existing single or multiple reservoirs, with no change in the volume of any of the reservoirs; or (b) The project activity is implemented in existing single or multiple reservoirs, where the volume of the reservoir(s) is increased and the power density, calculated using equation (7), is greater than 4 W/m^2; or	Not applicable to this project

Methodology ID	Applicability condition	Justification of compliance
	(c) The project activity results in new single or multiple reservoirs and the power density, calculated using equation (7), is greater than 4 W/m^2; or (d) The project activity is an integrated hydro power project involving multiple reservoirs, where the power density for any of the reservoirs, calculated using equation (7), is lower than or equal to 4 W/m2, and all of the following conditions shall apply: (i) The power density calculated using the total installed capacity of the integrated project, as per equation (8), is greater than 4 W/m^2; (ii) Water flow between reservoirs is not used by any other hydropower unit which is not a part of the project activity; (iii) Installed capacity of the power plant(s) with power density lower than or equal to 4 W/m2 are: a. Lower than or equal to 15 MW; and b. Less than 10 per cent of the total installed capacity of integrated hydro power project.	
	In the case of integrated hydro power projects, project participants shall: (a) Demonstrate that water flow from upstream power plants/units spill directly to the downstream reservoir and that collectively constitute to the generation capacity of the integrated hydro power project; or	Not applicable to this project

Methodology ID	Applicability condition	Justification of compliance
	(b) Provide an analysis of the water balance covering the water fed to power units, with all possible combinations of reservoirs and without the construction of reservoirs. The purpose of water balance is to demonstrate the requirement of specific combination of reservoirs constructed under CDM project activity for the optimization of power output. This demonstration has to be carried out in the specific scenario of water availability in different seasons to optimize the water flow at the inlet of power units. Therefore, this water balance will take into account seasonal flows from river, tributaries (if any), and rainfall for minimum of five years prior to the implementation of the CDM project activity.	
	In the case of PSP, the project participants shall demonstrate in the VCS Project Description that the project is not using water which would have been used to generate electricity in the baseline.	Not applicable to this project
	The methodology is not applicable to: (a) Project activities that involve switching from fossil fuels to renewable energy sources at the site of the project activity, since in this case the baseline may be the continued use of fossil fuels at the site; (b) Biomass fired power plants/units.	Not applicable to this project
	In the case of retrofits, rehabilitations, replacements, or capacity additions, this methodology is only applicable if	Not applicable to this project

Methodology ID	Applicability condition	Justification of compliance
	the most plausible baseline scenario, as a result of the identification of baseline scenario, is “the continuation of the current situation, that is to use the power generation equipment that was already in use prior to the implementation of the project activity and undertaking business as usual maintenance”.	

Methodology ID	Applicability condition	Justification of compliance
CDM TOOL 01	The use of the “Tool for the demonstration and assessment of additionality” is not mandatory for project participants when proposing new methodologies. Project participants may propose alternative methods to demonstrate additionality for consideration by the Executive Board. They may also submit revisions to approved methodologies using the additionality tool.	This project does not entail proposing a new methodology. As such TOOL 01 is mandatory, and therefore was applied to prove additionality.
CDM TOOL 01	Once the additionally tool is included in an approved methodology, its application by project participants using this methodology is mandatory.	TOOL 01 is referenced in the methodology ACM0002 as the prescribed tool to prove additionality and, as such it is mandatory for this project.

Methodology ID	Applicability condition	Justification of compliance
CDM TOOL 07	This tool may be applied to estimate the Operating Margin (OM), Build Margin (BM) and/or Combined Margin (CM) when calculating baseline emissions for a project activity that substitutes grid electricity that is where a project activity	This project will feed electricity into the South African national grid, and as such, this is the prescribed tool to calculate the baseline emissions.

	supplies electricity to a grid or a project activity that results in savings of electricity that would have been provided by the grid (e.g. demand-side energy efficiency projects).	
CDM TOOL 07	Under this tool, the emission factor for the project electricity system can be calculated either for grid power plants only or, as an option, can include off-grid power plants. In the latter case, two sub-options under step 2 of the tool are available to the project participants, i.e. option II a and option IIb. If option II a is chosen, the conditions specified in “Appendix 1: Procedures related to off-grid power generation” should be met. Namely, the total capacity of off-grid power plants (in MW) should be at least 10 per cent of the total capacity of grid power plants in the electricity system; or the total electricity generation by off-grid power plants (in MWh) should be at least 10 per cent of the total electricity generation by grid power plants in the electricity system; and that factors which negatively affect the reliability and stability of the grid are primarily due to constraints in generation and not to other aspects such as transmission capacity.	This project does not include any off-grid power plants. Therefore, this tool was used to calculate the emission factor for the project baseline electricity system for grid power plants only.
CDM TOOL 07	In case of CDM projects the tool is not applicable if the project electricity system is located partially or totally in an Annex I country.	The project activity is located in a non-Annex I country and hence the tool is applicable.
CDM TOOL 07	Under this tool, the value applied to the CO_2 emission factor of biofuels is zero.	The project activity does not involve biofuels.

Methodology ID	Applicability condition	Justification of compliance
CDM TOOL 32	The use of this methodological tool is not mandatory for the project participants of a CDM project activity or CDM PoA for demonstrating their additionality.	Noted
CDM TOOL 32	This methodological tool shall be applied in conjunction with a small-scale or large-scale methodology which refers to this tool.	This tool is applied in conjunction with large-scale ACM0002 methodology
CDM TOOL 32	The positive lists as contained in section 5 of this tool are valid up to 10 March 2025. Notwithstanding the provisions on the validity of new, revised and previous versions of methodologies and methodological tools in the “Procedure: Development, revision and clarification of baseline and monitoring methodologies and methodological tools”, there will be no grace period for the application of this tool and the validity of the positive list after this date, including in cases where further technologies are added to the positive list through revisions of this tool before this date	This tool is applicable at the date of entering validation for concentrating solar power technology for large-scale grid-connected power generation

3.3 Project Boundary

As per the methodology ACM002 version 22, the project boundary includes the power plant and all power plants connected physically to the electricity system that the project power plant is connected to.

Source		Gas	Included?	Justification/Explanation
Baseline	Grid Connected Electricity Generation	CO_2	Yes	Main emission source (Grid electricity)
		CH_4	No	Excluded for simplification. Minor source of emissions. Conservative
		N_2O	No	Excluded for simplification. Minor source of emissions. Conservative
		Other	No	No other GHG emissions relevant
Project	Concentrated Solar Plant Project Activity	CO_2	No	Excluded. Fossil fuels only used for emergency back-up generator and therefore taken as de minimis.
		CH_4	No	Not accounted as no or very little CH_4 emitted by the project activity
		N_2O	No	Not accounted as no or very little N_2O emitted by the project activity
		Other	No	No other GHG emissions

Figure 10 below shows the Redstone CSP plant under construction, with the neighbouring Jasper and Lesedi solar PV facilities. While the project boundary is the national electricity grid system shown in Figure 12 .

Figure 10: Redstone CSP Project under construction

The following diagram, Figure 11, shows the equipment, systems and flows of mass and energy, Including the GHG emission sources identified in the project boundary.

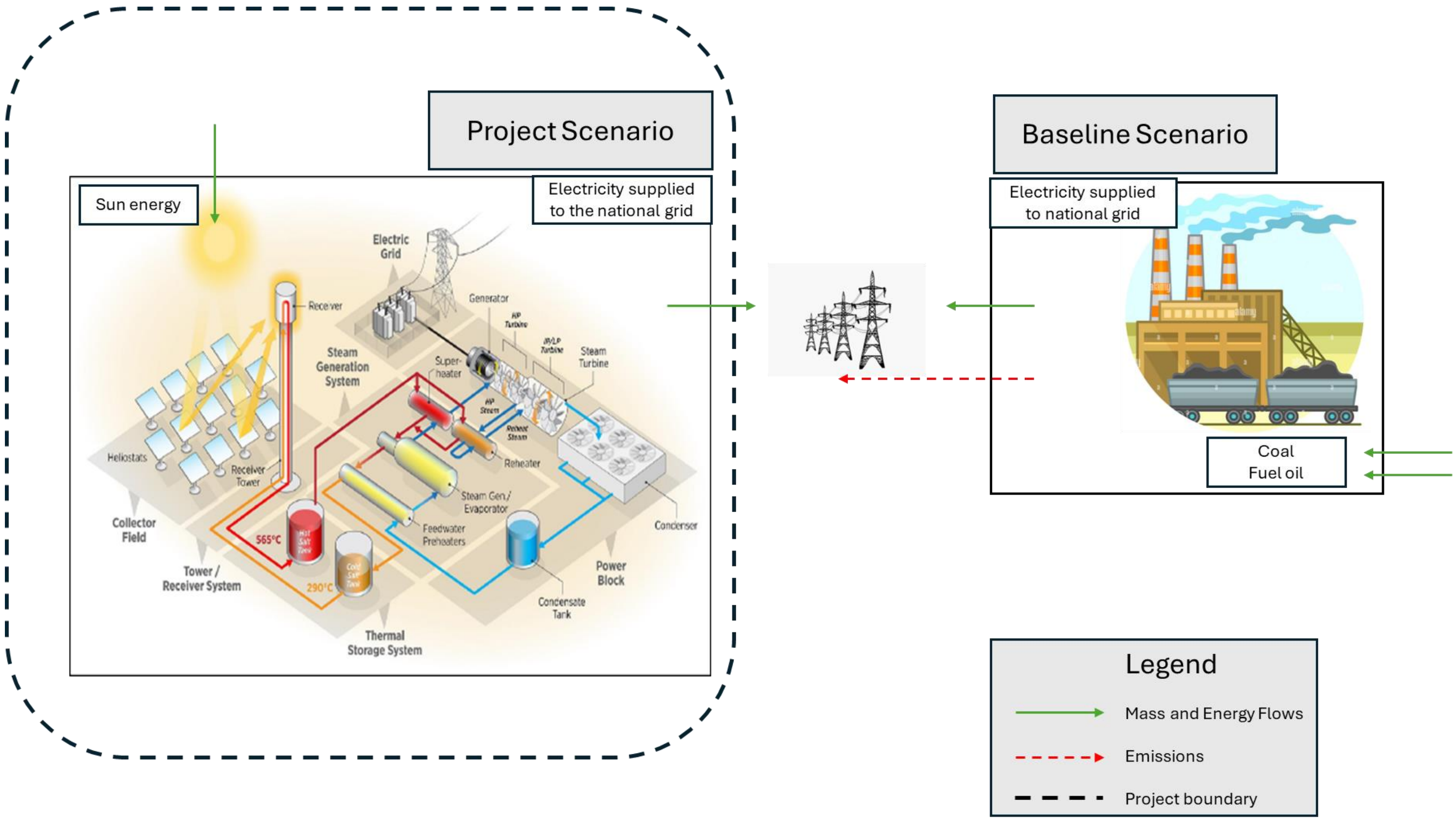

Figure 11: Project boundary, flows of mass and energy at Redstone CSP plant

3.4 Baseline Scenario

The baseline scenario for the Redstone project is defined as the continuation of current grid operations, where electricity demand is primarily met by existing grid-connected fossil fuel power plants, predominantly coal, and the addition of new fossil fuel-based generation sources. This approach aligns with the requirements of the ACM0002 (Version 22.0) methodology, which states in Section 5.2.1: "If the project activity is the installation of a Greenfield power plant with or without Battery Energy Storage System (BESS), the baseline scenario is electricity delivered to the grid by the project activity that would otherwise have been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in AM TOOL07 *'Tool to calculate the emission factor for an electricity system (Version 07.0)'."*

Justification of the Baseline Scenario:

The South African electricity grid is characterized by high carbon intensity due to its heavy reliance on coal-fired power plants, which accounts for 80% of its energy mix [11]. This high reliance on coal positions South Africa among the top greenhouse gas emitters per capita globally [3].

According to South Africa's Integrated Resource Plan (IRP 2019) [10], coal remains a dominant energy source, projected to account for over 50% of the electricity generation mix through 2030. The IRP outlines significant reliance on existing coal-fired power plants, with limited expansion of renewable energy capacity due to financial, regulatory, and infrastructural barriers. Despite government efforts to increase renewable energy through initiatives like the Renewable Energy Independent Power Producer Procurement Programme (REIPPPP), the expansion pace is slow, and integration challenges persist, resulting in continued reliance on fossil fuel generation.

The South African energy sector faces numerous challenges, including delays in project approvals, difficulties in securing financing, and grid connectivity issues. These barriers significantly impact the deployment of renewable energy projects, validating the assumption that the baseline scenario will continue to be dominated by fossil fuel-based power plants, including both existing and new coal and gas-fired units.

Historical data from Eskom, the primary electricity supplier, indicates a stable grid emission factor of approximately 0.959 $tCO2e/MWh$ over the past five years, driven by a heavy reliance on coal-fired plants. This trend is expected to continue in the absence of major interventions like the Redstone project, further supporting the selection of this baseline scenario.

The grid emission factor for this baseline is calculated in accordance with the CDM TOOL07 "Tool to calculate the emission factor for an electricity system" Version 07.0. Option A under the Simple OM approach was selected along with the default weighting for solar projects in the first crediting period. This results in a combined margin factor for the baseline grid of 0.959 tCO_2e/MWh.

The steps taken to calculate the grid emission factor as per AM Tool 07 are outlined in section 4.1 below.

Paragraphs 28 – 34 of ACM0002 do not apply to this project, as shown in *Table 7* below:

Table 7: Identification of Baseline scenario as per ACM0002

Section number in ACM0002	Applicable to	Applicability to the project
5.2.2	Capacity addition to an existing renewable energy power plant or integration of a BESS to an existing solar photovoltaic or wind power plant/unit	N/A – the project is a Greenfield facility and not a capacity addition to an existing renewable energy power plant.
5.2.3	Baseline scenario for retrofit or rehabilitation or replacement of an existing power plant	N/A – the project is a Greenfield facility and not a retrofit, rehabilitation or replacement of an existing power plant.
5.2.4	Baseline scenario for retrofit of an existing solar photovoltaic or wind power plant/unit with the integration of BESS	N/A – the project is a Greenfield facility and not a retrofit of an existing facility.
5.2.5	Baseline scenario for pumped storage projects operating in coordination with the renewable energy plants	N/A - the project is a Greenfield CSP facility and is not connected to a Greenfield pumped storage power plant.

The project activity is the installation of a greenfield power plant. The baseline scenario is electricity delivered to the grid by the project activity that would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in TOOL07.

3.5 Additionality

Additionality is demonstrated in the section below.

3.5.1 Regulatory Surplus

Is the project located in an UNFCCC Annex 1 or Non-Annex 1 country?

☐ Annex 1 country ☒ Non-Annex 1 country

Are the project activities mandated by any law, statute, or other regulatory framework?

☐ Yes ☒ No

If the project is located inside a Non-Annex 1 country and the project activities are mandated by a law, statute, or other regulatory framework, are such laws, statutes, or regulatory frameworks systematically enforced?

N/A – the project is located in the Republic of South Africa which is a Non-Annex 1 country and activities are not mandated by a law, statute, or other regulatory framework.

3.5.2 Additionality Methods

Based on the procedure outlined in ACM0002, the simplified procedure to demonstrate additionality was applied, as specified in CDM TOOL 32 (Positive lists of technologies v 04.0). As per paragraph 13 (§5.2.1 (a)), Solar thermal electricity generation including concentrating solar power appears on the positive list. It is therefore deemed additional if, at the time of the VCS Project Description submission, the percentage share of total installed capacity of the specific technology in the total installed grid connected power generation capacity in the host country is equal to or less than two per cent.

At the time of project submission, the total installed grid connected power generation capacity in South Africa was 53 898 MW [13]. The total installed CSP capacity in South Africa was 600 MW [1], which is 1,1% of the total installed capacity. The project is therefore deemed automatically additional.

3.6 Methodology Deviations

There are no methodology deviations in this project.

4 QUANTIFICATION OF ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

The baseline emissions were quantified in accordance with the methodology ACM0002 version 22.0. For this project, it includes electricity delivered to the South African National grid. This was calculated as set out in AM tool 07, and the calculation steps are described below.

Baseline Emission Calculation

Baseline emissions include only CO_2 emissions from electricity generation in fossil fuel fired power plants that are displaced due to the project activity. The methodology assumes that all project electricity generation above baseline levels would have been generated by existing grid-connected power plants and the addition of new grid-connected power plants. The baseline emissions are to be calculated as follows:

$$BE_y = EG_{PJ,y} \; x \; EF_{grid,CM,y}$$

Equation 1 (ACM0002 Equation 11)

Where:

BE_y = Baseline emissions in year y (tCO_2/yr)

$EG_{PJ,y}$ = Net electricity fed into the grid supplied by the project activity in year y (MWh/yr)

$EF_{grid,CM,y}$ = Combined margin CO_2 emission factor for grid connected power generation in year y calculated using TOOL07 (t CO_2/MWh)

The results obtained are: 479 416,66 t CO_2/ annum, as shown below:

Parameter name	Value	Unit
BE_y	479 416,66	t CO_2/ yr
$EG_{PJ,y}$	500 000,00	MWh/yr
$EF_{grid,CM,y}$	0,96	t CO_2 /MWh

The project is a greenfield power plant and therefore $EG_{PJ,y}$ is calculated as follows:

$$EG_{PJ,y} = EG_{facility,y}$$

Equation 2 (ACM0002 Equation 12)

Where:

$EG_{PJ,y}$	=	Net electricity generation fed into the grid as a result of the implementation of the project activity in year y (MWh/yr)
$EG_{facility,y}$	=	Net electricity generation supplied by the project plant to the grid in year y (MWh/yr)

The results obtained are: 500 000 MWh / yr (from technical specifications)

Grid Emission Factor Calculation

The grid emission factor for South Africa's national electricity grid was calculated as the weighted average of the system's operating margin (OM) and build margin (BM). The operating margin refers to the group of existing power plants whose current electricity generation would be affected by the proposed renewable energy project activity. The build margin refers to the group of prospective power plants whose construction and future operation would be affected by the proposed renewable energy project activity. This calculation is only applicable to grid-connected power plants and does not include off-grid power plants.

In accordance with Tool07 The following 6 steps are followed:

- **Step 1:** Identify the relevant electricity systems
- **Step 2:** Choose whether to include off-grid power plants in the project electricity system (optional)
- **Step 3:** Select a method to determine the operating margin (OM)
- **Step 4:** Calculate the operating margin emission factor according to the selected method
- **Step 5:** Calculate the build margin (BM) emission factor
- **Step 6:** Calculate the combined margin (CM) emission factor.

4.1.1 Step 1: Identify the project electricity system

The project electricity system was identified, using Option 2, as: The national grid of South Africa – administered by the national power utility, Eskom Holding SOC Ltd. This is shown schematically in Figure 12 below.

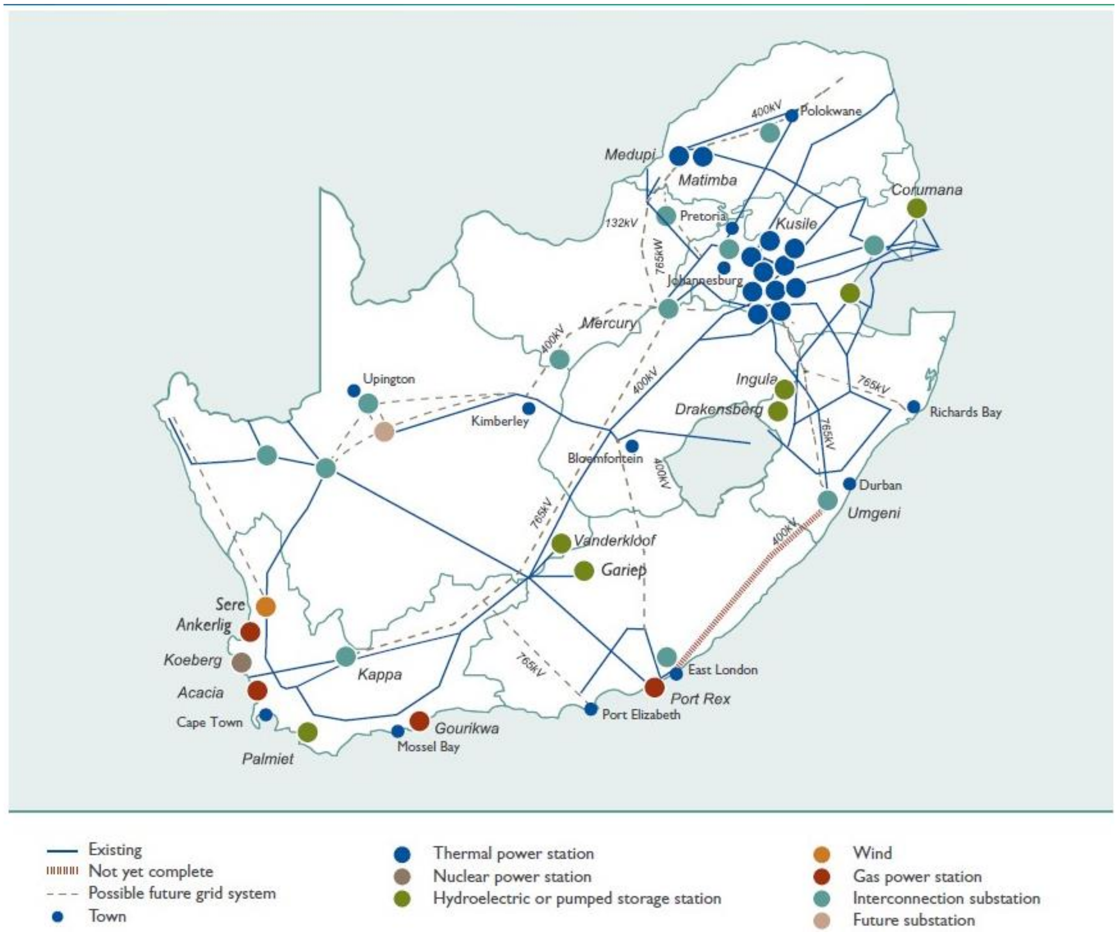

Figure 12: Project electricity system Source: [13]

South Africa's electricity grid is not connected to an electricity system located partially or totally in an Annex I country, therefore the grid emission factor is not zero.

4.1.2 Step 2: Choose whether to include off-grid power plants in the project electricity system

Only grid connected plants (shown in Table 10). were included in this calculation.

4.1.3 Step 3: Operating Margin method selection

The operating margin (OM) emission factor ($EF_{grid,OM,y}$) was calculated based on the simple OM method (paragraph 38.a in TOOL07). This was decided based on the following criteria:

- The share of electricity supplied by Low-Cost/Must-Run (LCMR) plants in the last 5 years has ranged from 19% – 23%. It has been consistently lower than 50%, as calculated using TOOL07 40(a) Approach 1, by Equation 3 below:

$$Share_{LCMR} = average\left[\frac{EG_{LCMR(y-4)}}{total_{y-4}}, \ldots, \frac{EG_{LCMR(y)}}{total_y}\right]$$

Equation 3 (Tool07, 40 Equation 1)

Where:

Share$_{LCMR}$	=	Share of the low cost/must run resources (%)
EG_{LCMRy}	=	Electricity generation supplied by LCMR plants in year y (MWh)
$total_y$	=	Total electricity supplied to the project electricity system in year y (MWh)
Y	=	The most recent year for which data is available

The ex-ante option (TOOL07 paragraph 42(a)) was chosen to determine the simple OM. The 3-year generation-weighted average emission factor was calculated based on the most recent data available at the time of listing the project on Verra's registry. For this project, the most recent data available is 2023. Therefore, the generation weighted average of 2021, 2022, and 2023 was used.

The results obtained are shown in Table 8 below

Table 8: LCMR Analysis

Parameter name	Value	Unit
Share$_{LCMR}$: 2023	23,1%	Per cent
Share$_{LCMR}$: 2022	21,3%	Per cent
Share$_{LCMR}$: 2021	19,4%	Per cent
EG_{LCMRy}: 2023	49 691	GWh / yr
EG_{LCMRy}: 2022	48 092	GWh / yr
EG_{LCMRy}: 2021	42 495	GWh / yr
$total_y$: 2023	215 318	GWh / yr
$total_y$: 2022	226 226	GWh / yr
$total_y$: 2021	219 423	GWh / yr

4.1.4 Step 4: Operating margin calculation

The simple OM emission factor was calculated as the generation-weighted average CO_2 emissions per unit net electricity generation (tCO_2/MWh) of all generating power plants serving the system, not including LCMR power plants. The following plants were identified as LCMR, and therefore excluded from the OM calculation:

Table 9 LCMR plants serving the project electricity system

LCMR Plant name	Technology	Reason for classification as LCMR
• Gariep • Vanderkloof • Mbashe • First Falls • Ncora • Second Falls	Hydroelectric power stations	Renewable (low generation costs)
• Drakensberg • Ingula • Palmiet	Pumped storage stations	Only used during periods of peak demand (must run)
• Acacia • Ankerlig • Gourikwa • Port Rex	Gas and liquid turbine peaking stations	Operated only during hours of highest daily, weekly or seasonal loads (must run)
• Sere	Wind energy stations	Renewable (low generation costs)
• Koeberg	Nuclear power stations	Nuclear listed as LCMR
• Biomass • Concentrated solar power • Gas/liquid fuel • Hydroelectric • Landfill • Solar PV • Wind	Imports and REIPPP plants	Imports or renewable

The power plants used in the OM calculation are shown in Table 10 below. The simple OM was calculated based on the net electricity generation and the CO_2 emission factor of each power unit (TOOL07 paragraph 47, option A), as per Equation 4.

$$EF_{grid,OMsimple,y} = \frac{\sum m\ EG_{m,y}\ \ x\ EF_{EL,m,y}}{\sum m\ EG_{m,y}}$$

Equation 4 (Tool 07 48, Equation 3)

Where:

$EF_{grid,OMsimple,y}$	=	Simple operating margin CO_2 emission factor in year y (t CO_2/MWh)
$EG_{m,y}$	=	Net electricity delivered to the grid by power unit m in year y (MWh)
$EF_{EL,m,y}$	=	CO_2 emission factor of power unit m in year y (t CO_2/MWh)
m	=	All power units serving the grid in year y except low-cost/must-run power units
y	=	The relevant year as per the data vintage chosen

The results obtained are: $EF_{grid,OMsimple,y}$ = 1.008 tCO_2/MWh
All values used are shown in Table 10.

Table 10: Operating Margin Plant Data

Plant Name	Electricity generated (MWh/yr) 2021	Electricity generated (MWh/yr) 2022	Electricity generated (MWh/yr) 2023	Emission factor 2023 (tCO_2/MWh)	Emission factor 2022 (tCO_2/MWh)	Emission factor 2021 (tCO_2/MWh)
Arnot	10 148 766	9 085 064	7 530 158	1,17	1,16	1,14
Duvha	11 230 400	9 822 489	9 555 266	1,11	1,13	1,13
Hendrina	4 873 506	4 738 179	2 136 620	1,35	1,24	1,20
Kendal	15 106 544	15 139 746	15 750 630	1,02	1,05	1,04
Kriel	11 328 488	12 597 519	10 727 877	1,32	1,28	1,22
Lethabo	21 832 979	22 890 481	21 367 191	1,00	0,97	0,95
Matimba	23 533 126	23 180 336	23 257 630	1,02	1,04	0,99
Majuba	21 857 835	19 823 301	15 540 628	1,19	1,13	1,09
Matla	18 560 297	18 412 664	15 170 449	1,30	1,27	0,99
Tutuka	10 554 111	8 524 715	7 564 337	1,18	1,23	1,22
Camden	2 518 438	6 670 341	7 378 745	1,10	1,16	1,25
Grootvlei	2 992 859	2 445 162	2 416 293	1,21	1,27	1,24
Komati	708 862	608 728	369 330	1,01	1,03	1,06
Kusile	8 553 383	7 992 189	9 119 476	1,01	1,30	1,09
Medupi	19 753 895	22 637 675	23 246 212	0,88	0,92	0,91
Total	**183 553 489**	**184 568 589**	**171 130 842**			

The emission factor of each power unit ($EF_{EL,m,y}$) was determined following option A1 (paragraph 49.a) as shown in Equation 5 below:

$$EF_{EL,m,y} = \frac{\sum_i FC_{i,m,y} \ x \ NCV_{i,y} \ x \ EF_{CO2,i,y}}{\sum EG_{m,y}}$$

Equation 5 (Tool 07 49.(a) Equation 4)

Where:

$EF_{EL,m,y}$	=	CO_2 emission factor of power unit m in year y (t CO_2/MWh)
$FC_{i,m,y}$	=	Amount of fuel type i consumed by power unit m in year y (Mass or volume unit)
$NCV_{i,y}$	=	Net calorific value of fuel type i in year y (GJ/mass or volume unit)
$EF_{CO2,i,y}$	=	CO_2 emission factor of fuel type i in year y (t CO_2/GJ)
$EG_{m,y}$	=	Net quantity of electricity delivered to the grid by power unit m in year y (MWh)
m	=	All power units serving the grid in year y except low-cost/must-run power units
i	=	All fuel types combusted in power unit m in year y
y	=	The relevant year as per the data vintage chosen

The emission factor for each power unit ($EF_{EL,m,y}$) is shown in Table 10 above.

The values used in Equation 5 are shown in Table 11 below.

$EG_{m,y}$ was obtained from Eskom's website as well as their annual Integrated Reports. [13], [18].

Table 11: Parameters for plant emission factor calculation

Plant Name	Coal usage (t) 2021	Coal usage (t) 2022	Coal usage (t) 2023	Fuel oil usage (t) 2023	Fuel oil usage (t) 2022	Fuel oil usage (t) 2021	Net calorific value coal (GJ/t)	Net calorific value fuel oil (GJ/t)	Emission factor coal (tCO_2/GJ)	Emission factor fuel oil (tCO_2/GJ)
Arnot	5 663 126	5 186 766	4 293 316	45 179,7	37 792,2	36 433,7	21,43	40,40	0,09	0,08
Duvha	6 581 000	5 732 195	5 428 210	71 179,4	39 550,9	34 435,1	20,39	40,40	0,09	0,08
Hendrina	3 024 527	3 024 981	1 486 626	7 722,1	11 012,5	7 013,8	20,51	40,40	0,09	0,08
Kendal	9 272 316	9 362 475	9 395 998	106 402,9	61 174,9	32 683,4	17,87	40,40	0,09	0,08
Kriel	6 694 760	7 838 445	6 865 127	53 949,1	39 516,2	47 877,9	21,64	40,40	0,09	0,08
Lethabo	15 158 029	16 277 939	15 646 686	14 075,3	12 792,0	9 740,7	14,53	40,40	0,09	0,08
Matimba	12 938 911	13 430 751	13 179 578	14 066,5	9 944,3	13 465,6	19,15	40,40	0,09	0,08
Majuba	12 773 092	11 924 693	9 831 151	140 169,2	149 002,7	89 746,1	19,58	40,40	0,09	0,08
Matla	10 694 976	10 842 908	9 536 046	15 923,0	14 520,0	9 179,0	18,19	40,40	0,09	0,08
Tutuka	6 697 281	5 510 214	4 697 013	64 592,8	68 779,3	124 521,3	19,83	40,40	0,09	0,08
Camden	1 569 620	3 932 224	4 129 344	49 016,8	57 014,4	41 321,3	20,45	40,40	0,09	0,08
Grootvlei	1 938 931	1 544 269	1 518 602	21 199,1	61 318,1	19 380,3	20,01	40,40	0,09	0,08
Komati	402 289	335 833	197 954	1 773,0	1 332,0	2 214,0	19,72	40,40	0,09	0,08
Kusile	5 047 912	4 783 403	5 035 930	10 351,0	534 043,2	32 165,0	19,42	40,40	0,09	0,08
Medupi	9 763 379	11 351 788	11 181 866	23 246,2	22 637,7	19 753,9	19,42	40,40	0,09	0,08

4.1.5 Step 5: Calculate the build margin (BM) emission factor

An ex-ante approach (option 1 in TOOL07 paragraph 72(a)) is selected to calculate the BM emission factor. The most recent information available on units already built for sample group m at the time of submitting the project for validation was used. The procedure followed to determine the sample group of power units (m) used to calculate the build margin is described below. The total electricity generation (AEG_{total}) of all power plants directly supplying the system in 2023 was 184 165,63 GWh.

The set of five power units, excluding power units registered as CDM project activities, that started to supply electricity to the grid most recently ($SET_{5\ units}$) was identified as shown in Table 12. The annual generation of $SET_{\underline{5\text{-}units}}$ was 35 365,31 GWh in 2023, and this set comprises 19,2% of AEG_{total}.

Table 12: Five most Recent Plants supplying the grid

No.	Plant name	Commissioning Date	Electricity supplied UOM	2023
1	Kusile	2017/08/30	GWh/yr	9119,48
2	Medupi	2015/08/22	GWh/yr	23246,21
3	Grootvlei	2008/04/30	GWh/yr	2416,29
4	Komati	2009/03/31	GWh/yr	369,33
5	Sere Wind	2015/03/01	GWh/yr	214
	TOTAL			**35 365,31**

The set of power units, excluding power units registered as CDM project activities, that started to supply electricity to the grid most recently and that comprise 20 per cent of AEG_{total} was determined next, and $SET_{\geq 20\%}$ generated 38 379,02 MWh as shown in Table 13 below, comprised 20,8% of AEG_{total}.

Table 13: Set of power plants that contribute 20% to the Eskom fleet power generation

No.	Plant name	Commissioning Date	UoM	Power Produced in Year: 2023
1	**Kusile**	**2017/08/30**	**GWh/yr**	**9119,48**
2	**Medupi**	**2015/08/22**	**GWh/yr**	**23246,21**
3	**Ankerlig**	**2007/03/01**	**GWh/yr**	**1332,49**
4	**Gourikwa**	**2007/07/01**	**GWh/yr**	**1681,22**
5	**Grootvlei**	**2008/04/30**	**GWh/yr**	**2416,29**
6	**Komati**	**2009/03/31**	**GWh/yr**	**369,33**
7	**Sere Wind**	**2015/03/01**	**GWh/yr**	**214,00**
	TOTAL			**38 379,02**

SET>20% comprised the larger annual electricity generation and therefore forms the sample set SETsample, but it did contain plants that were commissioned more than 10 years ago and therefore step (d) is followed. units that were commissioned more than 10 years ago were excluded from the set (those shaded grey in Table **13**), and CDM, Gold Standard (GS), and VCS projects were included until the electricity generation of the new set comprises 20% of the annual electricity generation of the project electricity system [19]. This resulted in a set that comprised 20% of the project electricity system and therefore steps (e) and (f) were not required. The resulting SETsample-CDM group of power units, m, used to calculate the build margin is shown

Table 14: Plant set for BM Calculation

No.	Plant name	Commissioning Date	UoM	Power Produced in Year: 2023
	Eskom Fleet Plants			
1	Kusile	2017/08/30	GWh/yr	9119,48
2	Medupi	2015/08/22	GWh/yr	23246,21
3	Sere Wind	2015/03/01	GWh/yr	214,00
	CDM, VCS and Gold Standard Plants			
4	Amakhala Emoyeni Grid Connected 138.6 MW Wind Farm, Phase 1	2013/03/01	GWh/yr	375,17
5	Karoo Renewable Energy Facility (Nobelsfontein Solar PV)	2014/04/01	GWh/yr	94,53
6	Bokpoort CSP (Concentrating Solar Power) Project, South Africa	2013/01/10	GWh/yr	225,86
7	Springbok Grid Connected 55.5 MW Wind Farm (Longyuan Mulilo)	2013/06/01	GWh/yr	130,30
8	Electricawinds 30 MW Wind Project at Riverbank	2013/12/01	GWh/yr	80,30
9	Longyuan Mulilo 1	2012/01/01	GWh/yr	308,14
10	Longyuan Mulilo 2 de Aar, 144 MW	2012/01/01	GWh/yr	497,38
11	Bokpoort CSP (Concentrating Solar Power) Project South Africa	2016/01/01	GWh/yr	200,00
12	Karoo Renewable Energy Facility (Nobelsfontein Wind)	2012/12/01	GWh/yr	878,44
13	Caledon Wind Farm in South Africa	2012/06/01	GWh/yr	406,01
14	Langa Energy Photovaltaic Solar Energy Facility	2012/03/05	GWh/yr	140,45
15	Indwe Wind Project	2012/06/01	GWh/yr	148,09
16	Grid Connected Wind Power Plant in Witberg	2012/02/01	GWh/yr	507,20
17	Red Cap Kouga Wind Farm	2011/11/02	GWh/yr	290,00
	TOTAL			**4281,85**

As the annual electricity generation of this set comprises at least 20% of the annual electricity generation of the project electricity system (i.e. $AEG_{SET\text{-}sample\text{-}CDM}$ > 0.2 x AEGtotal), this sample group ($SET_{sample\text{-}CDM}$) is used to calculate the build margin.

The build margin emission factor was calculated as the generation-weighted average emission factor (tCO_2/MWh) for power units m during 2023, which is the most recent year y for which electricity generation data is available, using Equation 6:

$$EF_{grid,BM,y} = \frac{\sum m\ EG_{m,y}\ \ x\ EF_{EL,m,y}}{\sum m\ EG_{m,y}}$$

Equation 6 (Tool 07 77 Equation 15)

Where:

$EF_{grid,BM,y}$	=	Build margin CO_2 emission factor in year y (tCO_2/MWh)
$EG_{m,y}$	=	Net quantity of electricity delivered to the grid by power unit m in year y (MWh)
$EF_{EL,m,y}$	=	CO_2 emission factor of power unit m in year y (tCO_2/MWh)
m	=	Power units included in the build margin
y	=	Most recent historical year for which electricity generation data is available

The results obtained are: $EF_{grid,BM,y}$ = 0.81
$EG_{m,y}$ and $EF_{EL,m,y}$ were used as shown in Table 10, $EG_{m,y}$ was used as shown in **Error! Reference source not found.**.

4.1.6 Step 6: Calculate the combined margin (CM) emission factor

The combined margin (CM) emission factor ($EF_{grid,CM,y}$) was calculated as the weighted average CM, as per Equation 7.

$$EF_{grid,CM,y} = EF_{grid,OM,y}\ x\ w_{OM} + EF_{grid,BM,y}\ x\ w_{BM}$$

Equation 7 (Tool 07 85 Equation 16)

As per TOOLI07, the following weighted values (for solar power generation) were used for w_{OM} and w_{BM}:

- w_{OM} = 0.75
- w_{BM} = 0.25

The results obtained are: $EF_{grid,CM,y}$ = 0,959

4.2 Project Emissions

Project emissions are calculated as follows, in accordance with ACM0002 version

$$PE_y = PE_{FF,y} + PE_{GP,y} + PE_{HP,y}$$

Equation 8 (ACM0002 Equation 1)

Where:

PE_y	=	Project emissions in year y (tCO_2)
$PE_{FF,y}$	=	Project emissions from fossil fuel consumption in year y (tCO_2)
$PE_{GP,y}$	=	Project emissions from the operation of geothermal power plants due to the release of non-condensable gases in year y (tCO_2 e/yr)
$PE_{HP,y}$	=	*Project* emissions from water reservoirs of hydro power plants (tCO_2 e/yr)

The project does not involve geothermal power plants or water reservoir of hydro power plants therefore the equation simplifies to:

$$PE_y = PE_{FF,y}$$

This plant will not use fossil fuels for electricity generation, only for emergency backup, therefore $PE_{FF,y}$ can be neglected, in accordance with ACM0002 paragraph 42. Thus project emissions are zero.

4.3 Leakage Emissions

No leakage emissions are considered. The emissions potentially arising due to activities such as power plant construction and upstream emissions from fossil fuel use (e.g. extraction, processing, transport etc.) are neglected, in accordance with ACM0002.

4.4 Estimated GHG Emission Reductions and Carbon Dioxide Removals

The project's emission reductions are calculated as follows:

$$ER_y = BE_y - PE_y$$

Equation 9 (ACM0002 Equation 17)

Where:

ER_y	=	Emission reductions in year y (t CO_2/yr)
BE_y	=	Baseline emissions in year y (t CO_2/yr)
PE_y	=	Project emissions in year y (t CO_2/yr)

Vintage period	Estimated baseline emissions (tCO_2e)	Estimated project emissions (tCO_2e)	Estimated leakage emissions (tCO_2e)	Estimated reduction VCUs (tCO_2e)	Estimated removal VCUs (tCO_2e)	Estimated total VCUs (tCO_2e)
25-Oct-2024 to 31-Dec-2024	125 000	0	0	125 000	0	125 000
01-Jan-2925 to 31-Dec-2025	500 000	0	0	500 000	0	500 000
01-Jan-2026 to 31-Dec-2026	500 000	0	0	500 000	0	500 000
01-Jan-2027 to 31-Dec-2027	500 000	0	0	500 000	0	500 000
01-Jan-2028 to 31-Dec-2028	500 000	0	0	500 000	0	500 000
01-Jan-2029 to 31-Dec-2029	500 000	0	0	500 000	0	500 000
01-Jan-2030 to 31-Dec-2030	500 000	0	0	500 000	0	500 000
01-Jan-2031 to 24-Oct-2031	375 000	0	0	375 000	0	375 000
Total	3 500 000	0	0	3 500 000	0	3 500 000

5 MONITORING

5.1 Data and Parameters Available at Validation

The table below provides all data and parameters that are determined or available at validation and remain fixed throughout the project crediting period.

Data / Parameter	$EF_{grid,OM,y}$
Data unit	tCO2/MWh
Description	Simple operating margin CO_2 emission factor in year y
Source of data	Calculated using Equation 1, as per TOOL07 "Tool to calculate the emission factor for an electricity system" Data Sources: Records published on Eskom's website monthly for each power station to fulfil the Atmospheric emission license (EAL) requirements. Eskom's integrated annual report for 2023
Value applied	1.008
Justification of choice of data or description of measurement methods and procedures applied	The simple OM emission factor was calculated as the generation-weighted average CO_2 emissions per unit net electricity generation (tCO_2/MWh) of all generating power plants serving the system, not including low-cost/must-run power plants/units. This means that only Eskom's coal-based power plants were included in the calculation. The OM was calculated as per option A (paragraph 47.a) of TOOL07 "Tool to calculate the emission factor for an electricity system", based on the net electricity generation and a CO_2 emission factor of each power unit.
Purpose of data	Calculation of baseline emissions
Comments	No additional comments

Data / Parameter	$EF_{grid,BM,y}$
Data unit	tCO_2/MWh
Description	Build margin CO_2 emission factor in year y

Source of data	Calculated using Equation 6, as per TOOL07 "Tool to calculate the emission factor for an electricity system" Data source:
Value applied	0.811
Justification of choice of data or description of measurement methods and procedures applied	The BM was calculated as per option 1 (paragraph 72) of TOOL07 "Tool to calculate the emission factor for an electricity system". An ex-ante approach was followed, based on the most recent information available on units already built. The BM will be updated for the second crediting period based on the most recent information available at that time. Capacity additions from retrofits of power plants are not included in the calculation of the build margin emission factor.
Purpose of data	Calculation of baseline emissions
Comments	No additional comments

Data / Parameter	$EF_{grid,CM,y}$
Data unit	tCO_2/MWh
Description	Combine margin CO_2 emission factor in year y
Source of data	Calculated using Equation 7, as per TOOL07 "Tool to calculate the emission factor for an electricity system" Data Source:
Value applied	0,959
Justification of choice of data or description of measurement methods and procedures applied	The combined margin (CM) emission factor ($EF_{grid,CM,y}$) was calculated based on the weighted average CM method (paragraph 81.a) since the data to determine the BM is available. This was done as per equation 16 of TOOL07 "Tool to calculate the emission factor for an electricity system". Where W_{OM} is equal to: 0.75 and W_{BM} is equal to: 0.25, for the first crediting period and for subsequent crediting periods
Purpose of data	Calculation of baseline emissions
Comments	No additional comments

Data / Parameter	Share_{LCMR}
Data unit	Percent (%)
Description	Share of the low cost/must run resources
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [13] and records published on Eskom's website [18] for each power station.
Value applied	Annual LCMR (%) <table><tr><td>2021</td><td>2022</td><td>2023</td></tr><tr><td>19,4%</td><td>21,3%</td><td>23,1%</td></tr></table>
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table 8
Purpose of data	To calculate baseline emissions, using Equation 3.
Comments	The simple operating margin (OM) can be used where low-cost/must-run resources constitute less than 50 per cent of total grid generation (excluding electricity generated by off-grid power plants) in the average of the five most recent years, determined by using Equation 1 under Approach 1 from section 40 in Tool 07.

Data / Parameter	$\text{EG}_{LCMR,y}$
Data unit	MWh
Description	Electricity generation supplied to the project electricity system by the low cost/must run resources in year *y*
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [13]and records published on Eskom's website [20] for each power station.
Value applied	<table><tr><td>$EG_{LCMRy:}$ 2023</td><td>49 691 000</td></tr><tr><td>$EG_{LCMRy:}$ 2022</td><td>48 092 000</td></tr><tr><td>$EG_{LCMRy:}$ 2021</td><td>42 495 000</td></tr></table>
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table 8

Purpose of data	To calculate baseline emissions, using Equation 3.
Comments	The simple operating margin (OM) can be used where low-cost/must-run resources constitute less than 50 per cent of total grid generation (excluding electricity generated by off-grid power plants) in the average of the five most recent years, determined by using Equation 1 under Approach 1 from section 40 in Tool 07.

Data / Parameter	total_y
Data unit	MWh
Description	Total electricity generation supplied to the project electricity system in year *y*
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [14] and monthly records published on Eskom's website [11] for each power station to fulfil the Atmospheric emission license (EAL) requirements. Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [13] and records published on Eskom's website [18] for each power station
Value applied	$total_y$: 2023 | 215 318 | GWh / yr $total_y$: 2022 | 226 226 | GWh / yr $total_y$: 2021 | 219 423 | GWh / yr $total_y$: 2023 | 215 318 000 $total_y$: 2022 | 226 226 000 $total_y$: 2021 | 219 423 000
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table 8As presented in Table 8
Purpose of data	To calculate baseline emissions, using Equation 3.
Comments	The simple operating margin (OM) can be used where low-cost/must-run resources constitute less than 50 per cent of total grid generation (excluding electricity generated by off-grid power plants) in the average of the five most recent years, determined by using Equation 1 under Approach 1 from section 40 in Tool 07.

Data / Parameter	Y
Data unit	Year
Description	The most recent year for which data is available
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [14] and monthly records published on Eskom's website [14] for each power station to fulfil the Atmospheric emission license (EAL) requirements. Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [13] and records published on Eskom's website [18] for each power station
Value applied	2023
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table 8
Purpose of data	To calculate baseline emissions, using Equation 3.
Comments	The simple operating margin (OM) can be used where low-cost/must-run resources constitute less than 50 per cent of total grid generation (excluding electricity generated by off-grid power plants) in the average of the five most recent years, determined by using Equation 1 under Approach 1 from section 40 in Tool 07.

Data / Parameter	$EF_{grid,OMsimple,y}$
Data unit	tCO_2e/MWh
Description	Simple operating margin CO_2 emission factor in year y
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [11] and monthly records published on Eskom's website [14] for each power station to fulfil the Atmospheric emission license (EAL) requirements. Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [13] and records published on Eskom's website [18] for each power station
Value applied	1,.008
Justification of choice of data or description of	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from

measurement methods and procedures applied	Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table 10
Purpose of data	To calculate baseline emissions, using Equation 4.
Comments	The simple OM emission factor is calculated as the generation-weighted average CO_2 emissions per unit net electricity generation (t CO_2/MWh) of all generating power plants serving the system, not including low-cost/must-run power plants. using Equation 3 under Option A from section 48 in Tool 07.

Data / Parameter	$\mathrm{EG}_{m,y}$
Data unit	MWh
Description	Net electricity delivered to the grid by power unit m in year y
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [11] and monthly records published on Eskom's website [14] for each power station to fulfil the Atmospheric emission license (EAL) requirements. Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [13] and records published on Eskom's website [18] for each power station
Value applied	As per Table 10

Plant Name	Electricity generated (MWh/yr) 2021	Electricity generated (MWh/yr) 2022	Electricity generated (MWh/yr) 2023
Arnot	10 148 766	9 085 064	7 530 158
Duvha	11 230 400	9 822 489	9 555 266
Hendrina	4 873 506	4 738 179	2 136 620
Kendal	15 106 544	15 139 746	15 750 630
Kriel	11 328 488	12 597 519	10 727 877
Lethabo	21 832 979	22 890 481	21 367 191
Matimba	23 533 126	23 180 336	23 257 630
Majuba	21 857 835	19 823 301	15 540 628
Matla	18 560 297	18 412 664	15 170 449
Tutuka	10 554 111	8 524 715	7 564 337

Camden	2 518 438	6 670 341	7 378 745
Grootvlei	2 992 859	2 445 162	2 416 293
Komati	708 862	608 728	369 330
Kusile	8 553 383	7 992 189	9 119 476
Medupi	19 753 895	22 637 675	23 246 212
Total	**183 553 489**	**184 568 589**	**171 130 842**

Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table 10
Purpose of data	To calculate baseline emissions, using Equation 4.
Comments	The generation-weighted average CO_2 emissions per unit net electricity generation (t CO_2/MWh) of all generating power plants serving the system, not including low-cost/must-run power plants is used to determine the simple OM emission factor, using Equation 3 under Option A from section 48 in Tool 07.

Data / Parameter	$\mathrm{EF}_{EL,m,y}$
Data unit	tCO_2/MWh
Description	CO_2 emission factor of power unit m in year y
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [11] and monthly records published on Eskom's website [14] for each power station to fulfil the Atmospheric emission license (EAL) requirements. Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [13] and monthly and annual records published on Eskom's website [20] for each power station to fulfil the Atmospheric emission license (EAL) requirements.
Value applied	As per Table 10

	Plant Name	Emission factor 2023 (tCO_2/MWh)	Emission factor 2022 (tCO_2/MWh)	Emission factor 2021 (tCO_2/MWh)
	Arnot	1,17	1,16	1,14
	Duvha	1,11	1,13	1,13
	Hendrina	1,35	1,24	1,20
	Kendal	1,02	1,05	1,04
	Kriel	1,32	1,28	1,22
	Lethabo	1,00	0,97	0,95
	Matimba	1,02	1,04	0,99
	Majuba	1,19	1,13	1,09
	Matla	1,30	1,27	0,99
	Tutuka	1,18	1,23	1,22
	Camden	1,10	1,16	1,25
	Grootvlei	1,21	1,27	1,24
	Komati	1,01	1,03	1,06
	Kusile	1,01	1,30	1,09
	Medupi	0,88	0,92	0,91
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The power plants m are shown in Table 10).			
Purpose of data	To calculate baseline emissions, using Equation 5.			
Comments	The data on fuel consumption and electricity generation is available for the selected m power plants, so Equation 4 under Option A1 from paragraph 49(a) in Tool 07 is applied.			

Data / Parameter	$FC_{i,m,y}$
Data unit	Tonnes
Description	Amount of fuel type i consumed by power unit m in year y

Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [13] and monthly and annual records published on Eskom's website [20] for each power station to fulfil the Atmospheric emission license (EAL) requirements	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [11] and monthly records published on Eskom's website [14] for each power station to fulfil the Atmospheric emission license (EAL) requirements. Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [13] and monthly and annual records published on Eskom's website [20] for each power station to fulfil the Atmospheric emission license (EAL) requirements
Value applied	See table below. As per Table 11

Plant Name	Coal usage (t) 2021	Coal usage (t) 2022	Coal usage (t) 2023	Fuel oil usage (t) 2023	Fuel oil usage (t) 2022	Fuel oil usage (t) 2021
Arnot	5 663 126	5 186 766	4 293 316	45 179,7	37 792,2	36 433,7
Duvha	6 581 000	5 732 195	5 428 210	71 179,4	39 550,9	34 435,1
Hendrina	3 024 527	3 024 981	1 486 626	7 722,1	11 012,5	7 013,8
Kendal	9 272 316	9 362 475	9 395 998	106 402,9	61 174,9	32 683,4
Kriel	6 694 760	7 838 445	6 865 127	53 949,1	39 516,2	47 877,9
Lethabo	15 158 029	16 277 939	15 646 686	14 075,3	12 792,0	9 740,7
Matimba	12 938 911	13 430 751	13 179 578	14 066,5	9 944,3	13 465,6
Majuba	12 773 092	11 924 693	9 831 151	140 169,2	149 002,7	89 746,1
Matla	10 694 976	10 842 908	9 536 046	15 923,0	14 520,0	9 179,0
Tutuka	6 697 281	5 510 214	4 697 013	64 592,8	68 779,3	124 521,3
Camden	1 569 620	3 932 224	4 129 344	49 016,8	57 014,4	41 321,3
Grootvlei	1 938 931	1 544 269	1 518 602	21 199,1	61 318,1	19 380,3
Komati	402 289	335 833	197 954	1 773,0	1 332,0	2 214,0
Kusile	5 047 912	4 783 403	5 035 930	10 351,0	534 043,2	32 165,0
Medupi	9 763 379	11 351 788	11 181 866	23 246,2	22 637,7	19 753,9

As per Table 11

Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The power plants m are shown in Table 11
Purpose of data	To calculate baseline emissions, using Equation 5.
Comments	The data on fuel consumption and electricity generation is available for the selected m power plants, so Equation 4 under Option A1 from paragraph 49(a) in Tool 07 is applied.

Data / Parameter	$NCV_{i,y}$
Data unit	GJ/tonne
Description	Net calorific value of fuel type i in year y
Source of data	DFFE for coal [21] IPCC default factors for heavy fuel oil , [22]
Value applied	For coal as per plant in Table 11

Plant Name	Net calorific value coal (GJ/t)	Net calorific value fuel oil (GJ/t)
Arnot	21,43	40,40
Duvha	20,39	40,40
Hendrina	20,51	40,40
Kendal	17,87	40,40
Kriel	21,64	40,40
Lethabo	14,53	40,40
Matimba	19,15	40,40
Majuba	19,58	40,40
Matla	18,19	40,40
Tutuka	19,83	40,40
Camden	20,45	40,40
Grootvlei	20,01	40,40
Komati	19,72	40,40
Kusile	19,42	40,40
Medupi	19,42	40,40

Justification of choice of data or description of measurement methods and procedures applied	Tier 2 NCV (for South Africa) only available for coal. Tier 1 used for other NCVs
Purpose of data	To calculate baseline emissions, using Equation 5.
Comments	The data on fuel consumption and electricity generation is available for the selected m power plants, so Equation 4 under Option A1 from paragraph 49(a) in Tool 07 is applied.

Data / Parameter	AEG_{total}
Data unit	GWh
Description	Total electricity generation of all power plants directly supplying the system in year y
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [11] and monthly records published on Eskom's website [14] for each power station to fulfil the Atmospheric emission license (EAL) requirements. Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [13] and monthly and annual records published on Eskom's website [20] for each power station to fulfil the Atmospheric emission license (EAL)
Value applied	184 165,63
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table 10.
Purpose of data	To calculate baseline emissions
Comments	

Data / Parameter	$SET_{5\ units}$
Data unit	Dimensionless
Description	The set of five power units, excluding power units registered as CDM project activities, that started to supply electricity to the grid most recently, and that comprise 20% of AEG_{total}
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [11] and monthly records published on Eskom's website [14] for each power station to fulfil

	the Atmospheric emission license (EAL) requirements. Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [13] and monthly and annual records published on Eskom's website [20] for each power station to fulfil the Atmospheric emission license (EAL)
Value applied	Kusile (2017/08/30) Medupi (2015/08/22) Sere Wind (2015/03/01) Komati (2009/03/31) Grootvlei (2008/04/40)
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table 12
Purpose of data	To calculate baseline emissions
Comments	This set of power units generate 20,8% of $\mathrm{AEG_{total}}$ in 2023.

Data / Parameter	$AEG_{set-5-units}$
Data unit	GWh
Description	Annual electricity generation of the set of five power units, excluding power units registered as CDM project activities, that started to supply electricity to the grid most recently, and that comprise 20% of AEG_{total}
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [11] and monthly records published on Eskom's website [14] for each power station to fulfil the Atmospheric emission license (EAL) requirements. Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [13] and monthly and annual records published on Eskom's website [20] for each power station to fulfil the Atmospheric emission license (EAL)
Value applied	35 365,31
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table 12
Purpose of data	To calculate baseline emissions
Comments	This set of power units generated 19,2% of $\mathrm{AEG_{total}}$ in 2023.

Data / Parameter	$SET_{\geq 20\%}$
Data unit	Dimensionless
Description	The set of power units, excluding power units registered as CDM project activities, and that comprise 20% of AEG_{total}
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [11] and monthly records published on Eskom's website [4] for each power station to fulfil the Atmospheric emission license (EAL) requirements. Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [13] and monthly and annual records published on Eskom's website [20] for each power station to fulfil the Atmospheric emission license (EAL)
Value applied	Kusile (2017/08/30) Medupi (2015/08/22) Ankerlig (2007/03/01) Gourikwa (2007/07/01) Grootvlei (2008/04/30) Komati (2009/03/31) Sere Wind (2015/03/01)
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table 13
Purpose of data	To calculate baseline emissions
Comments	This set of power units generate 20,8% of $\mathrm{AEG_{total}}$ in 2023.

Data / Parameter	$AEG_{set\geq 20\%}$
Data unit	GWh
Description	Annual electricity generation of the set of power units, excluding power units registered as CDM project activities, and excluding those commissioned more than 10 years ago, and that comprise 20% of AEG_{total}
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [11] and monthly records published on Eskom's website [14] for each power station to fulfil the Atmospheric emission license (EAL) requirements. Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [13] and monthly and annual records published on Eskom's website [20] for each power station to fulfil the Atmospheric emission license (EAL)

Value applied	38 379,02
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table 13
Purpose of data	To calculate baseline emissions
Comments	This set of power units generated 20,8% of $\mathrm{AEG}_{\mathrm{total}}$ in 2023.

Data / Parameter	SET_{sample}
Data unit	Dimensionless
Description	The set of power units, (between $SET_{5\ \mathrm{units}}$ and $SET_{\geq 20\%}$) that comprises the larger annual electricity generation, and exclude power units that started to supply electricity to the grid more than 10 years ago
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [11] and monthly records published on Eskom's website [14] for each power station to fulfil the Atmospheric emission license (EAL) requirements. Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [13] and monthly and annual records published on Eskom's website [20] for each power station to fulfil the Atmospheric emission license (EAL)
Value applied	Kusile (2017/08/30) Medupi (2015/08/22) Sere Wind (2015/03/01)
Justification of choice of data or description of measurement methods and procedures applied	**Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in** Table 14Table 13
Purpose of data	To calculate baseline emissions
Comments	This set of power units generated 32,579,69 GWh in 2023, comprising 17,7% of $\mathrm{AEG}_{\mathrm{total}}$ in 2023.

Data / Parameter	$AEG_{set\ sample}$
Data unit	GWh

Description	Annual electricity generation of the set of power units, excluding power units registered as CDM project activities, and excluding those commissioned more than 10 years ago, and that comprise 20% of AEG_{total}
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [11] and monthly records published on Eskom's website [14] for each power station to fulfil the Atmospheric emission license (EAL) requirements. Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [13] and monthly and annual records published on Eskom's website [20] for each power station to fulfil the Atmospheric emission license (EAL)
Value applied	32,579,69
Justification of choice of data or description of measurement methods and procedures applied	**Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in** Table 14Table 13
Purpose of data	To calculate baseline emissions
Comments	This set of power units generated >20% of $\mathrm{AEG_{total}}$ in 2023.

Data / Parameter	$SET_{\mathrm{sample-CDM}}$
Data unit	Dimensionless
Description	The set of power units, (between $SET_{5\ \mathrm{units}}$ and $SET_{\geq 20\%}$) that comprises the larger annual electricity generation, excludes power units that started producing electricity more than 10 years ago, and including power units registered as CDM project activities, that comprise 20% of AEG_{total}
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [11] and monthly records published on Eskom's website [14] for each power station to fulfil the Atmospheric emission license (EAL) requirements. Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [13] and monthly and annual records published on Eskom's website [20] for each power station to fulfil the Atmospheric emission license (EAL)
Value applied	Shown in Error! Reference source not found..
Justification of choice of data or description of	**Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from**

measurement methods and procedures applied	**Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in** **Table 14** Table 14
Purpose of data	To calculate baseline emissions
Comments	This set of power units generated 36 861,54 GWh in 2023, comprising 20,0% of AEG_{totla}

Data / Parameter	$AEG_{SET-sample-CDM}$
Data unit	GWh
Description	Annual electricity generation of the set of power units, excluding power units registered as CDM project activities, and excluding those commissioned more than 10 years ago, and that comprise 20% of AEG_{total}
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [11] and monthly records published on Eskom's website [14] for each power station to fulfil the Atmospheric emission license (EAL) requirements. Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [13] and monthly and annual records published on Eskom's website [20] for each power station to fulfil the Atmospheric emission license (EAL)
Value applied	36 861,54
Justification of choice of data or description of measurement methods and procedures applied	**Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in** **Table 14** Table 14
Purpose of data	To calculate baseline emissions
Comments	This set of power units generated ≥20,0% of $\mathrm{AEG_{total}}$ in 2023.
Data / Parameter	w_{OM}
Data unit	Percentage (%)
Description	Weighting of operating margin emissions factor

Source of data	Tool07 paragraph 86 (a)
Value applied	75%
Justification of choice of data or description of measurement methods and procedures applied	Default value provided in the CDM Tool 07.
Purpose of data	To calculate baseline emissions, using Equation 7
Comments	Owing to the intermittent and non-dispatchable nature of solar power generation, the weighting of the operating margin emission factor is set at 75% for the first crediting period and for subsequent crediting periods.

Data / Parameter	w_{BM}
Data unit	Percentage (%)
Description	Weighting of build margin emissions factor
Source of data	Tool07 paragraph 86 (a)
Value applied	25%
Justification of choice of data or description of measurement methods and procedures applied	Default value provided in the CDM Tool 07.
Purpose of data	To calculate baseline emissions, using Equation 7
Comments	Owing to the intermittent and non-dispatchable nature of solar power generation, the weighting of the build margin emission factor is set at 25% for the first crediting period and for subsequent crediting periods.

Data / Parameter	BE_y
Data unit	tCO_2e/yr
Description	Emission reductions in year y
Source of data	Quantified using Equation 1
Value applied	To be determined ex-post500 000

Justification of choice of data or description of measurement methods and procedures applied	Equation provided in ACM0002.
Purpose of data	To calculate baseline emissions, using Equation 1
Comments	

5.2 Data and Parameters Monitored

The table below provides all data and parameters that will be monitored during the project crediting period. The values provided are used to quantify the estimated reductions and removals for the project crediting period in Section 4 above.

Data / Parameter	$EG_{facility,y}$
Data unit	MWh/Year
Description	Quantity of net electricity generation supplied by the project (Solar) plant to the grid in year y
Source of data	Electricity reading meters
Description of measurement methods and procedures to be applied	Electricity generated is metered automatically, recorded electronically on the DCS and SCADA systems, recorded monthly, following the Plant's standard operating procedures, and checked against the Eskom meter (located at the independent Noko switching station).
Frequency of monitoring/recording	Continuous monitoring and monthly reporting.
Value applied	Ex ante values for the first crediting period: 25 October-2024 to 31-December-2024: 125 000 01-January-2025 to 31-December-2025: 500 000 01-January-2026 to 31-December-2026: 500 000 01-January-2027 to 31-December-2027: 500 000 01-January-2028 to 31-December-2028: 500 000

	01-January-2029 to 31-December-2029 — 500 000 01-January-2030 to 31-December-2030 — 500 000 01-January-2030 to 24 October- -2031 — 375 000 Total — 3 500 000
Monitoring equipment	Electricity meter, supplied by HLC and Emerson. Emerson Distributed Control System (DCS), Steam turbine controlled via a Siemens control system, which is integrated with the Emerson components as part of an overall Supervisory Control and Data Acquisition (SCADA) system.
QA/QC procedures to be applied	Electricity production *measure by Plant equipment, is corroborated monthly* against the Eskom meter readings (located at the independent Noko switching station). Any discrepancies will immediately trigger fault finding and corrective action procedures. There is also a backup meter both on site and at the Noko switching station which is used as an additional failsafe for QA/QC purposes. All reported electricity generation values will be checked by both Redstone personnel and Eskom personnel prior to billing and payment. The two electrical meters on site will be calibrated in accordance with the manufacturer's specifications and form part of the planned maintenance schedule for both checking and calibration
Purpose of data	Calculation of project emissions
Calculation method	Net electricity supplied to the grid by the project plant is directly measured and reported, not calculated.
Comments	No additional comments

Data / Parameter	PE_y
Data unit	tCO_2/yr
Description	Project emissions in year y
Source of data	Quantified using Euation 8
Description of measurement methods and procedures to be applied	Quantified, not measured
Frequency of monitoring/recording	Monthly

Value applied	0
Monitoring equipment	Quantified, not monitored
QA/QC procedures to be applied	Describe the quality assurance and quality control (QA/QC) procedures to be applied, including the calibration procedures where applicable.
Purpose of data	Calculation of project emissions
Calculation method	Equation 8
Comments	For all renewable energy power generation project activities, emissions due to the use of fossil fuels for the backup generator can be neglected. This project does not involve geothermal power plants or water reservoir of hydro power plants. Therefore project emissions for this renewable energy power generation project will be zero.

Data / Parameter	ER_y
Data unit	tCO_2/yr
Description	Emission reductions in year y
Source of data	Quantified using Equation 9
Description of measurement methods and procedures to be applied	Quantified, not measured
Frequency of monitoring/recording	Monthly
Value applied	To be calculated ex-post500 000
Monitoring equipment	Quantified, not monitored
QA/QC procedures to be applied	Describe the quality assurance and quality control (QA/QC) procedures to be applied, including the calibration procedures where applicable.
Purpose of data	Calculation of project emissions
Calculation method	Equation 9
Comments	The emission reductions determined at validation apply estimates of likely emission reductions from the proposed project activity during the crediting period, using available ex

	ante values. These will be revised during each verification using the actual data recorded (where applicable).

5.3 Monitoring Plan

Methods for measuring, recording, storing, aggregating, collating and reporting on monitored data and parameters.

Electricity generated by the CSP facility will be monitored internally using an Emerson Distributed Control System (DCS). The steam turbines will be controlled via a Siemens control system, which is integrated with the Emerson components as part of an overall Supervisory Control and Data Acquisition (SCADA) system which will be used for operational control and reporting. Apart from reporting for this project, the data will also be used for billing the national utility for electricity supplied to the grid. All data and reports will be stored and backed up in accordance with the SCADA system design. An example of an indicative DCS screen for the electricity generation and transfer is shown in Figure 13 below.

Measurement data is managed by the project's DCS. The measuring equipment is supplied by HLC and Emerson, and is installed in three steps:

- Calibration under standard temperature and pressure conditions
- Installation and electrical connection by a certified external contractor
- Commissioning and final test under project conditions by the internal instrumentation team

All data is stored on a server on-site and also loaded onto a firewall protected cloud storage system. All measuring equipment will strictly comply with their specified maintenance plans and calibration certificates are kept on record.

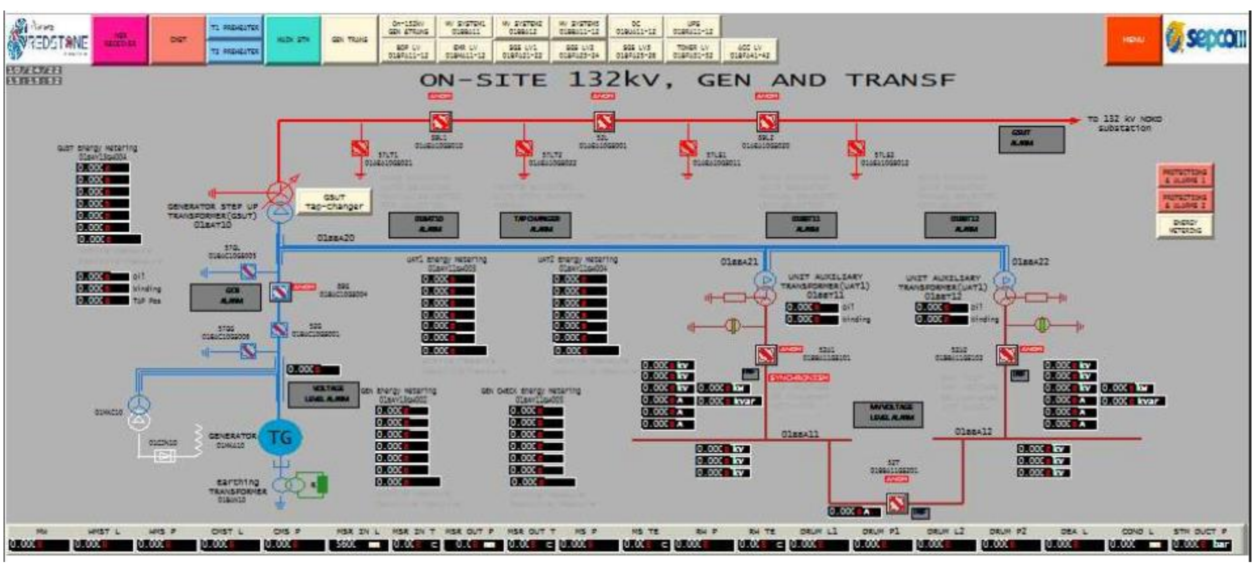

Figure 13: Indicative DCS screen for electricity generation and transfer

As per the applied methodology, electricity generation will be measured onsite at the outgoing substation, adjacent to the generating turbines. This will be monitored on a continuous basis, with a monthly report being aggregated and reported upon. A check meter, owned by the national utility (Eskom), is located at the Eskom Noko 132 kV Switching Station on the perimeter fence which serves as the grid tie-in point. Meters at both locations have a back-up meter installed as a failsafe.

The two electrical meters on site, calibrated in accordance with the manufacturer's specifications, will form part of the planned maintenance schedule for both checking and calibration. Instrument accuracy checks and verification will be conducted by Competent Instrumentation technicians and records kept on site. Calibrations will be conducted by competent technicians under SANAS accreditation. The applicable national standard to be applied is SANS 474 [23]. This defines the required frequency of calibration in §4.7.4 for meters greater than 10MVA as being every 5 years and less than 10MVA as being every 10 years.

Organizational structure, responsibilities and competencies of the personnel that will be carrying out monitoring activities.

Data collection, consolidation and results analysis will be undertaken by a dedicated data and instrumentation team who are adequately trained and competent. They will be responsible for the maintenance (planned and unplanned), checking and arranging calibration for plant equipment. They will also be responsible for maintaining records of calibrations and procurement of any instrumentation equipment that may be required for the operation of the plant. All instrument technicians will be suitably competent and records of their training and qualifications will be maintained by the Human Resources department. Only SANAS qualified organisations will conduct calibrations.

Procedures for internal auditing and QA/QC

Metered electricity production will be recorded by the plant on the DCS system. This will be checked against the Eskom meter at the independent Noko switching station. Any discrepancies will immediately trigger fault finding and corrective action procedures. There is also a backup meter both on site and at the Noko switching station which are used as an additional failsafe for QA/QC purposes. All reported electricity generation values will be checked by both Redstone personnel and Eskom personnel prior to billing and payment.

APPENDIX 1: COMMERCIALLY SENSITIVE INFORMATION

No commercially sensitive information is presented in this document

APPENDIX 2: BIBLIOGRAPHY

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